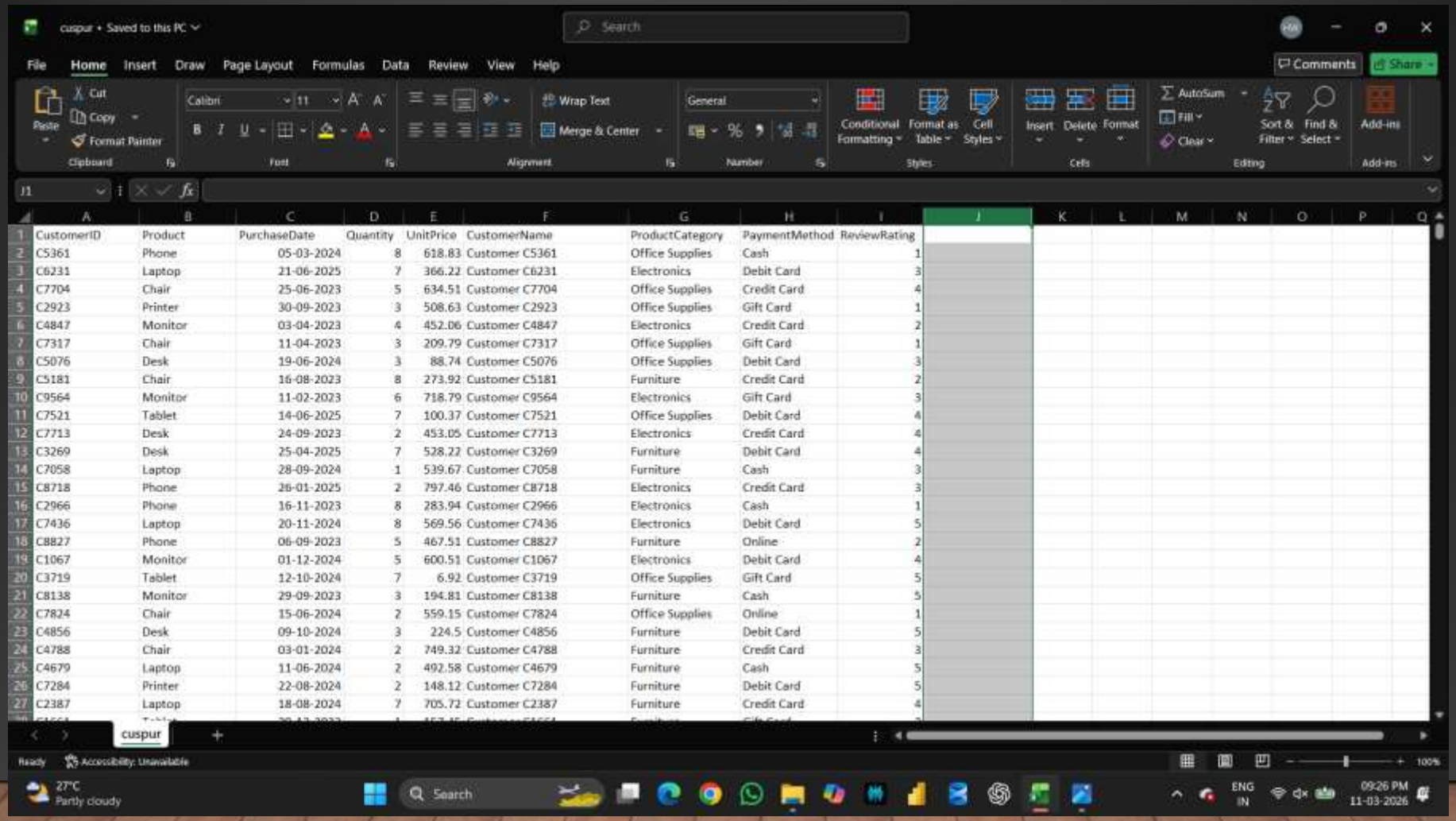
AZURE MINI PROJECT - 4

MULTI - SOURCE DATA INTEGRATION

PROJECT SUMMARY

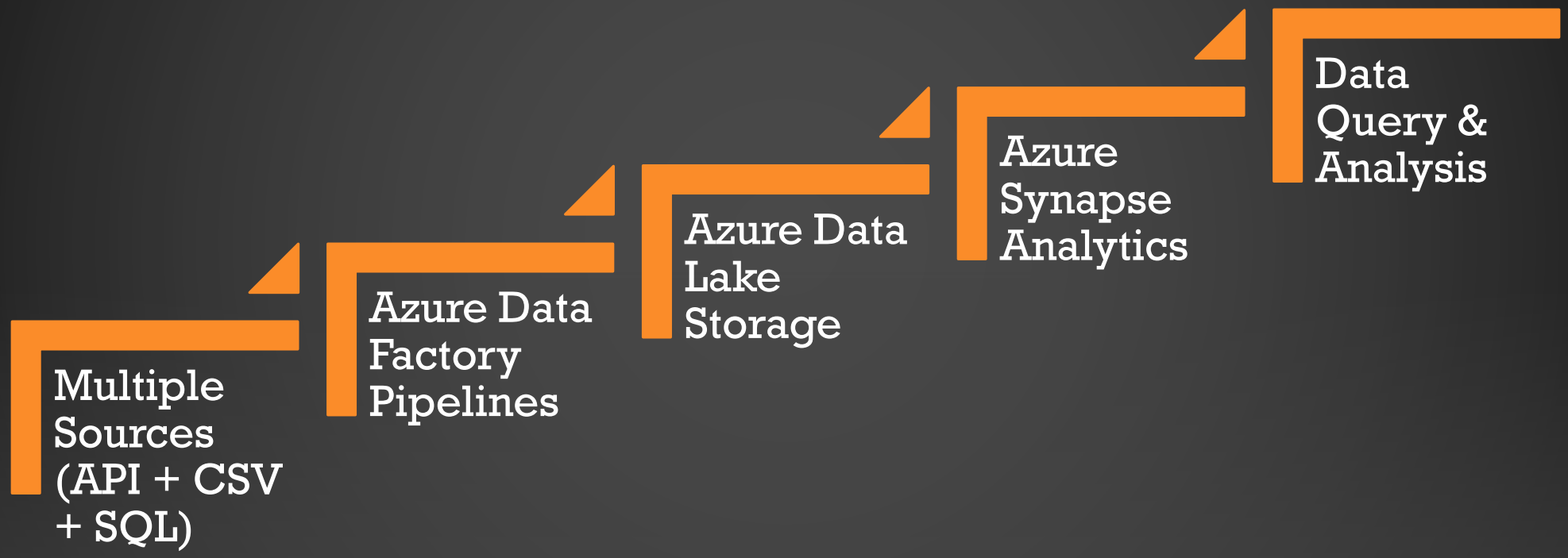
- Built a **Data Integration Pipeline** to collect data from multiple sources.
- Integrated **REST API, CSV files,** and **Azure SQL Database**.
- Used **Azure Data Factory** to automate data ingestion.
- Stored all raw data in **Azure Data Lake Storage**.
- Queried and analyzed data using **Azure Synapse Analytics (Serverless SQL)**..

CREATE A XL-SHEET FOR THE INPUT

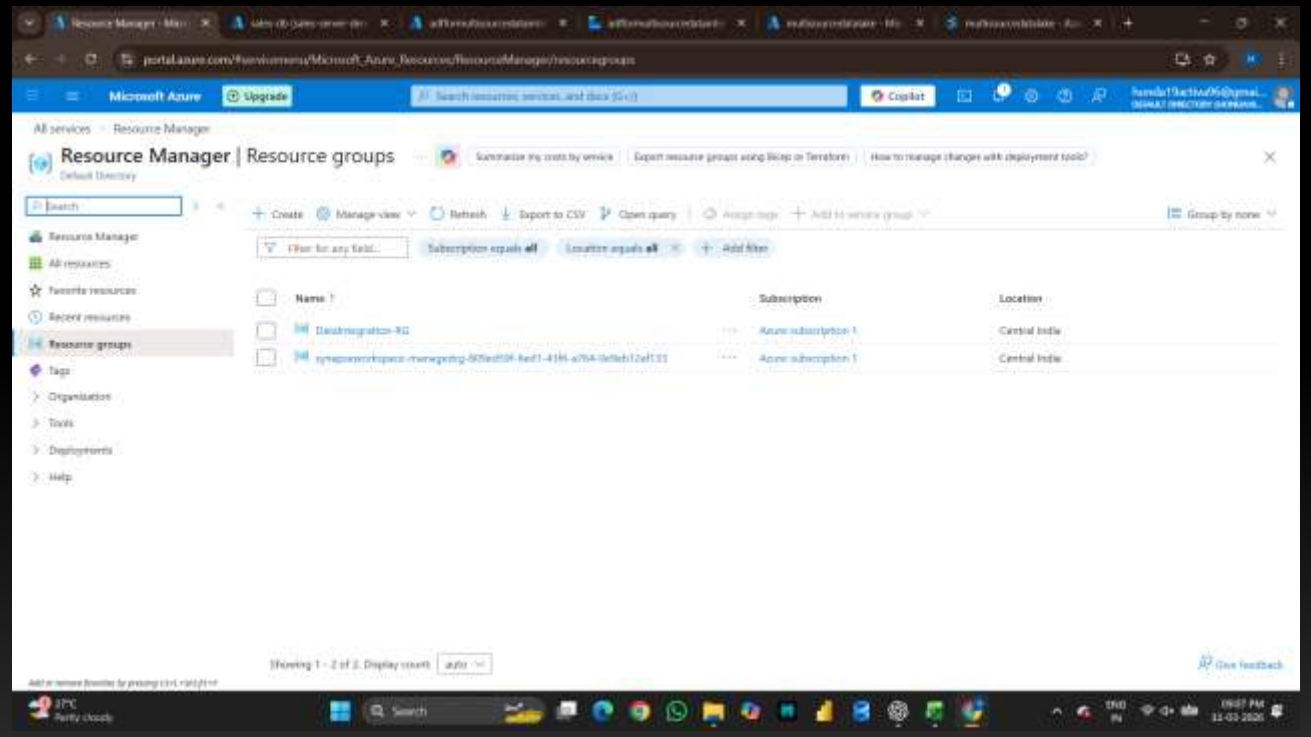


CustomerID	Product	PurchaseDate	Quantity	UnitPrice	CustomerName	ProductCategory	PaymentMethod	ReviewRating
C5361	Phone	05-03-2024	8	618.83	Customer C5361	Office Supplies	Cash	1
C6231	Laptop	21-06-2025	7	366.22	Customer C6231	Electronics	Debit Card	3
C7704	Chair	25-06-2023	5	634.51	Customer C7704	Office Supplies	Credit Card	4
C2923	Printer	30-09-2023	3	508.63	Customer C2923	Office Supplies	Gift Card	1
C4847	Monitor	03-04-2023	4	452.06	Customer C4847	Electronics	Credit Card	2
C7317	Chair	11-04-2023	3	209.79	Customer C7317	Office Supplies	Gift Card	1
C5076	Desk	19-06-2024	3	88.74	Customer C5076	Office Supplies	Debit Card	3
C5181	Chair	16-08-2023	8	273.92	Customer C5181	Furniture	Credit Card	2
C9564	Monitor	11-02-2023	6	718.79	Customer C9564	Electronics	Gift Card	3
C7521	Tablet	14-06-2025	7	100.37	Customer C7521	Office Supplies	Debit Card	4
C7713	Desk	24-09-2023	2	453.05	Customer C7713	Electronics	Credit Card	4
C3269	Desk	25-04-2025	7	528.22	Customer C3269	Furniture	Debit Card	4
C7058	Laptop	28-09-2024	1	539.67	Customer C7058	Furniture	Cash	3
C8718	Phone	26-01-2025	2	797.46	Customer C8718	Electronics	Credit Card	3
C2966	Phone	16-11-2023	8	283.94	Customer C2966	Electronics	Cash	1
C7436	Laptop	20-11-2024	8	569.56	Customer C7436	Electronics	Debit Card	5
C8827	Phone	06-09-2023	5	467.51	Customer C8827	Furniture	Online	2
C1067	Monitor	01-12-2024	5	600.51	Customer C1067	Electronics	Debit Card	4
C3719	Tablet	12-10-2024	7	6.92	Customer C3719	Office Supplies	Gift Card	5
C8138	Monitor	29-09-2023	3	194.81	Customer C8138	Furniture	Cash	5
C7824	Chair	15-06-2024	2	559.15	Customer C7824	Office Supplies	Online	1
C4856	Desk	09-10-2024	3	224.5	Customer C4856	Furniture	Debit Card	5
C4788	Chair	03-01-2024	2	749.32	Customer C4788	Furniture	Credit Card	3
C4679	Laptop	11-06-2024	2	492.58	Customer C4679	Furniture	Cash	5
C7284	Printer	22-08-2024	2	148.12	Customer C7284	Furniture	Debit Card	5
C2387	Laptop	18-08-2024	7	705.72	Customer C2387	Furniture	Credit Card	4

ARCHITECTURE DIAGRAM

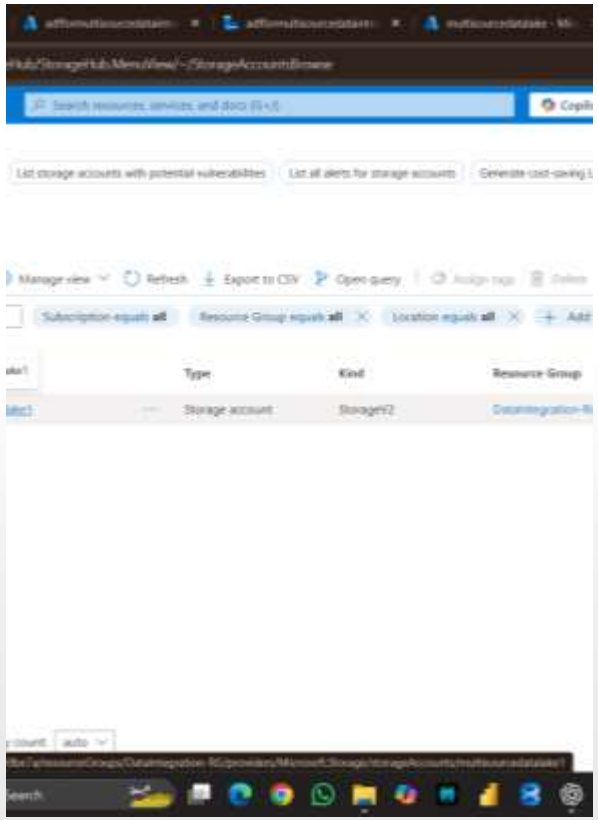


CREATE A RESOURCE GROUP

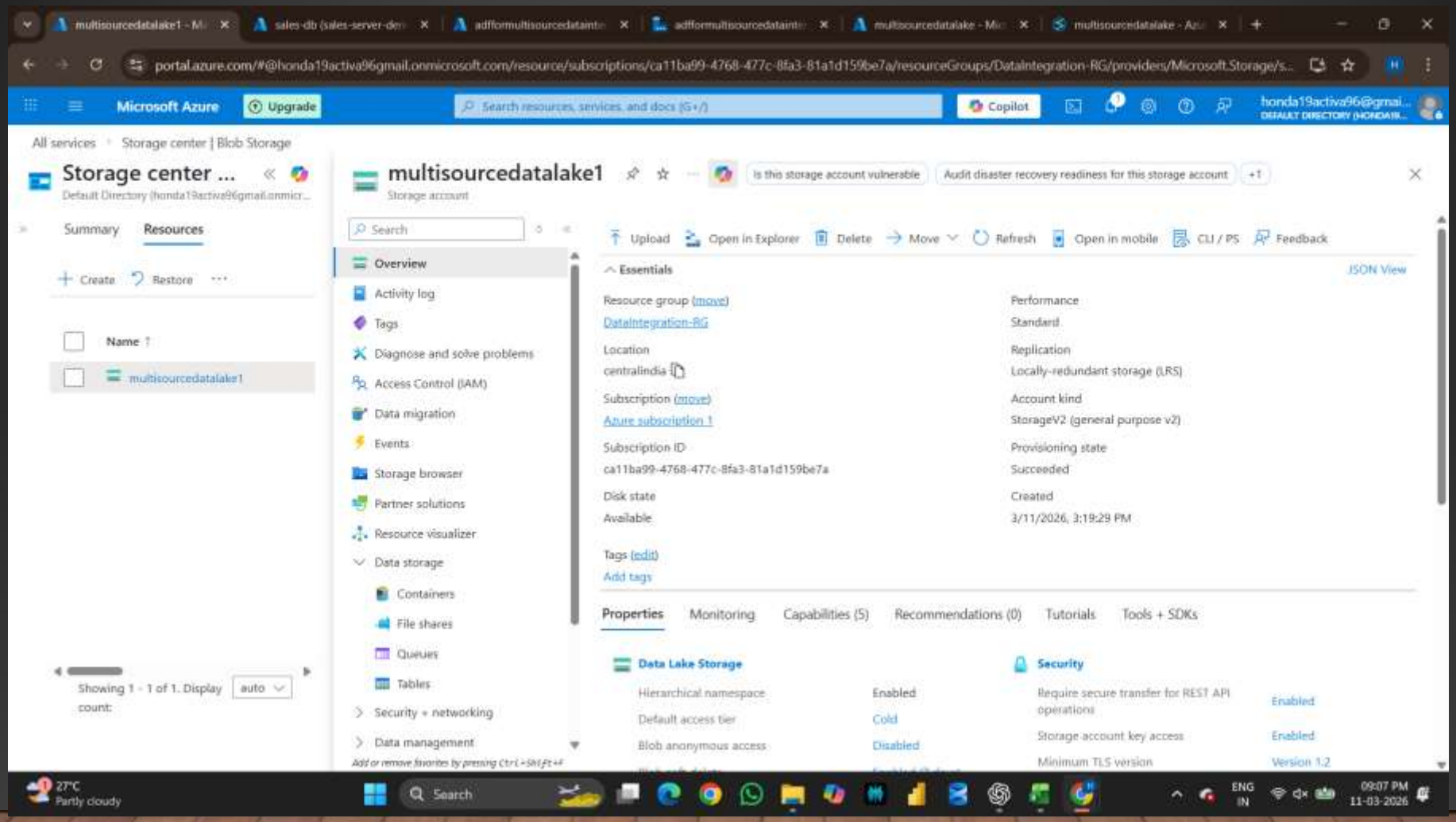


SAMPLE IMAGE FOR CREATING A STORAGE ACCOUNT

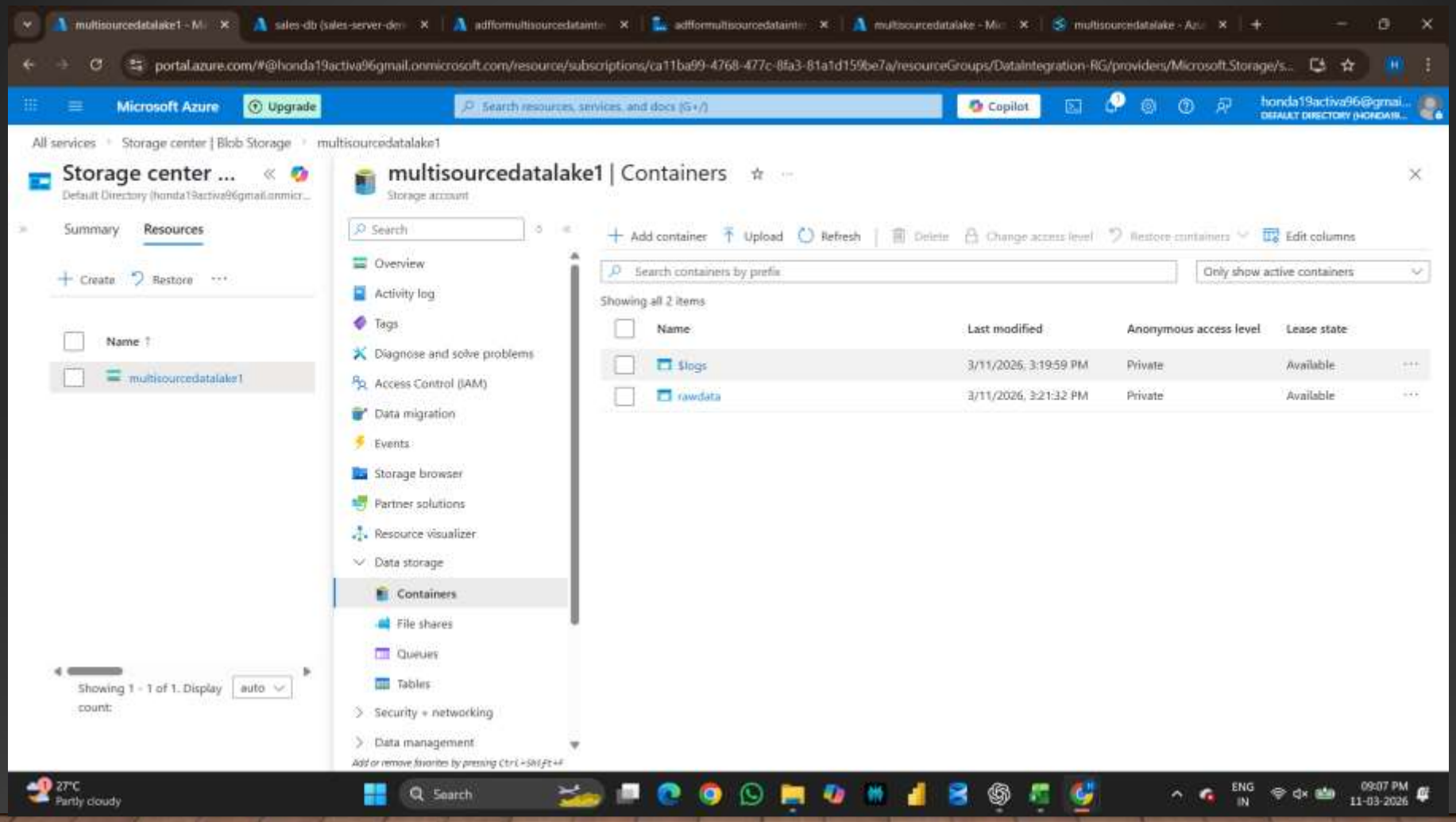
CREATE A STORAGE ACCOUNT



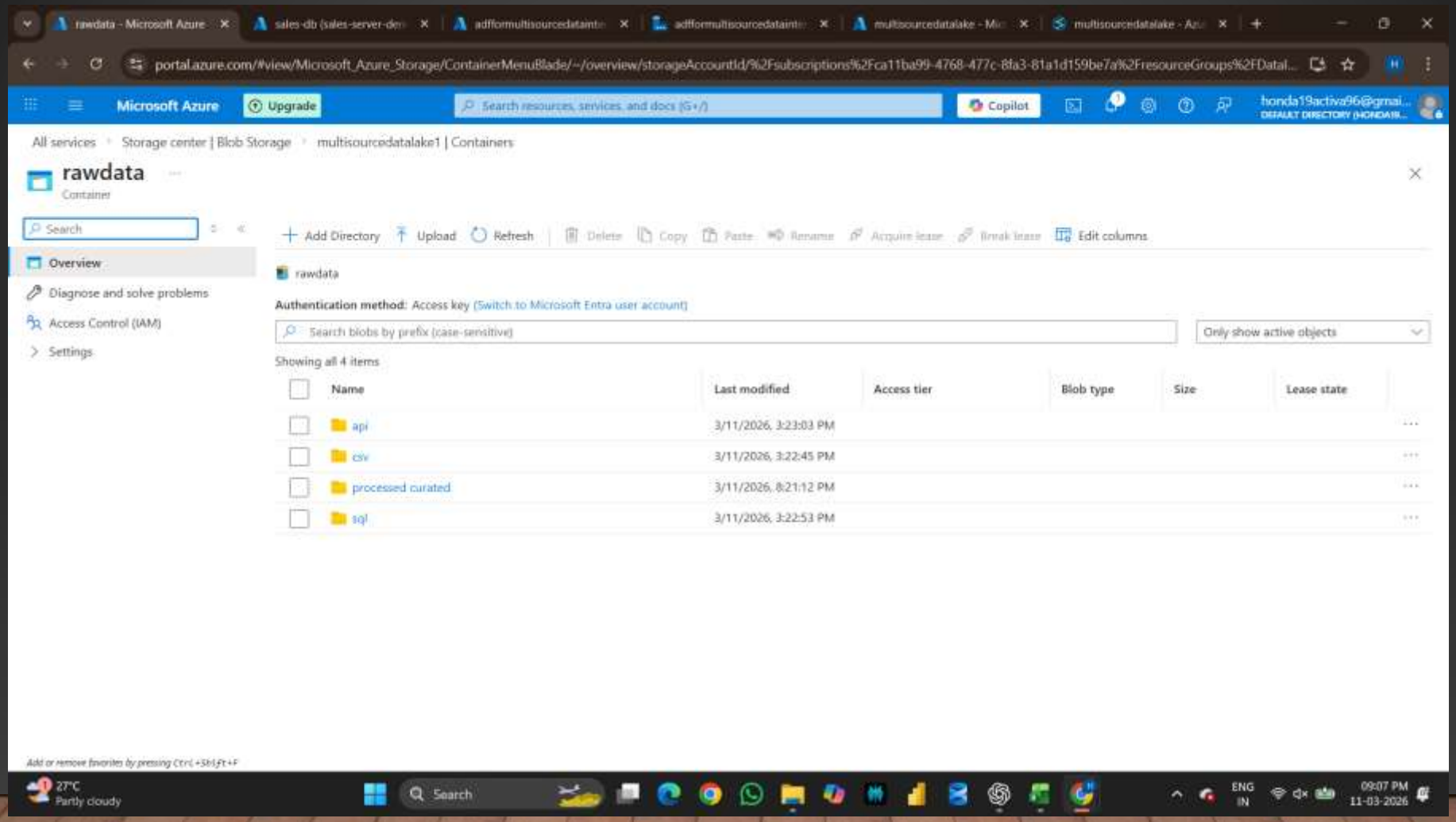
SAMPLE IMAGE - CREATING A STORAGE ACCOUNT - OVERVIEW



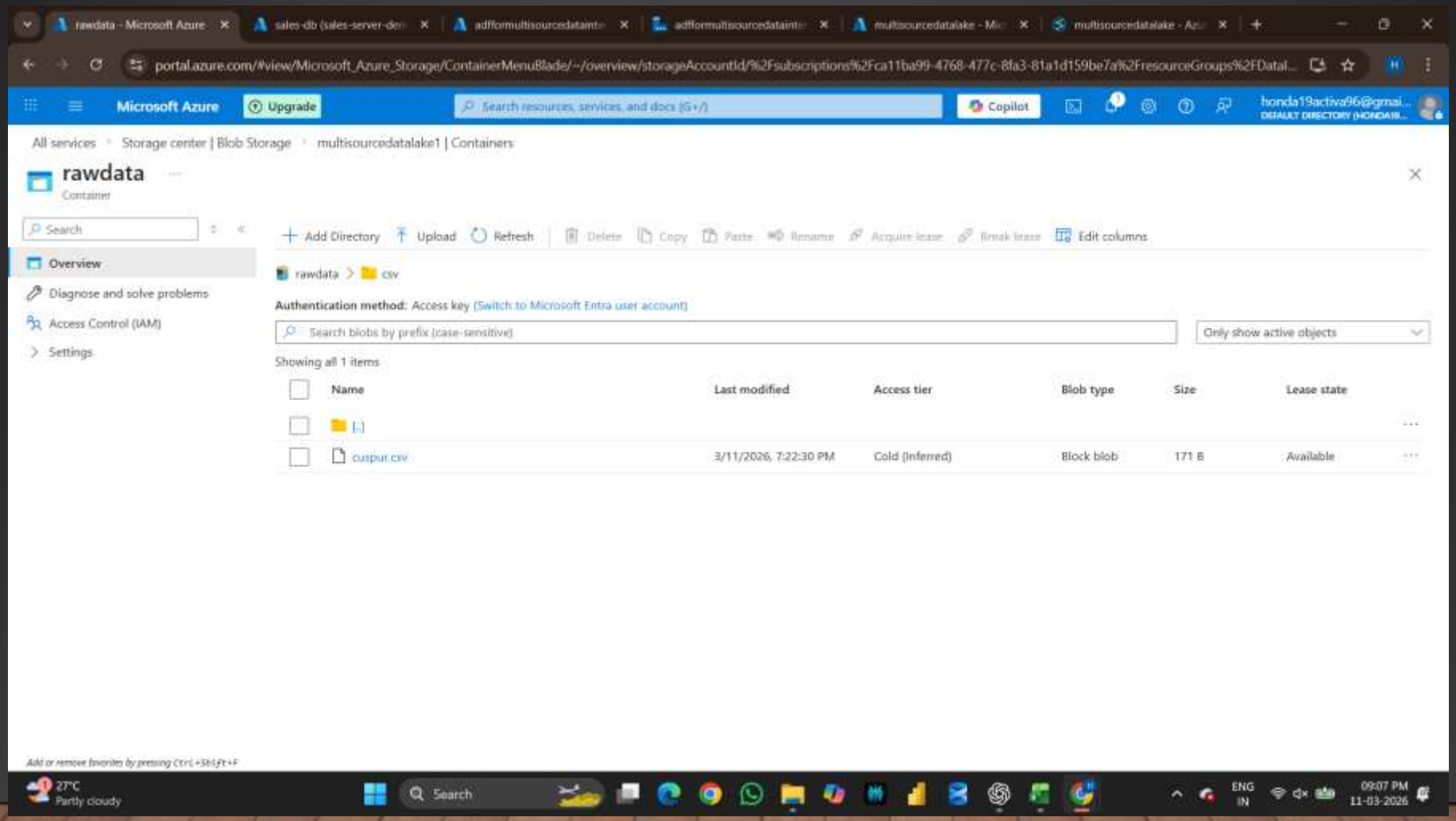
INSIDE A CONTAINER – CREATE ONE FOLDER (RAWDATA)



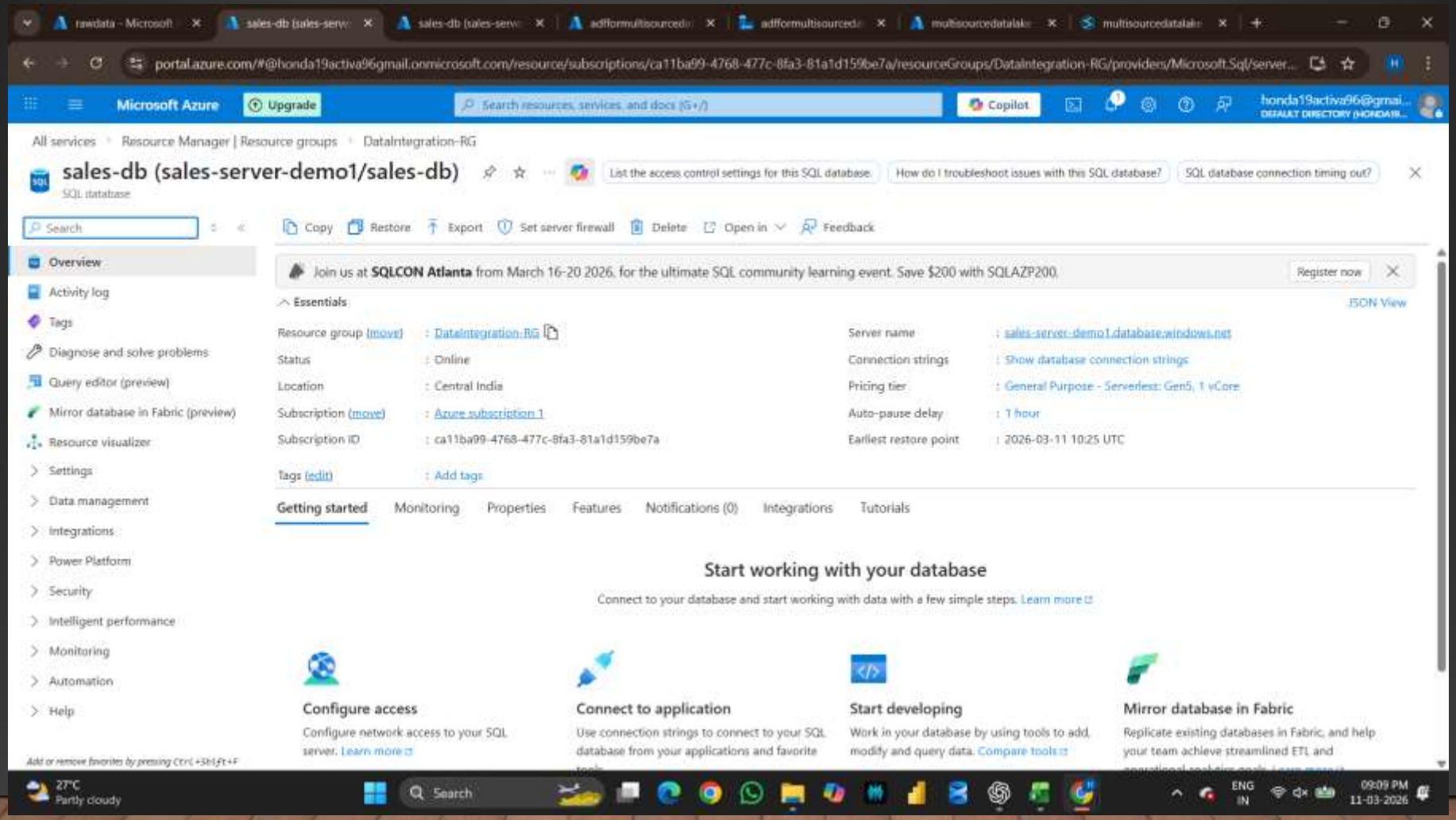
INSIDE A FOLDER CREATE A API , CSV , SQL (SUB FOLDERS)



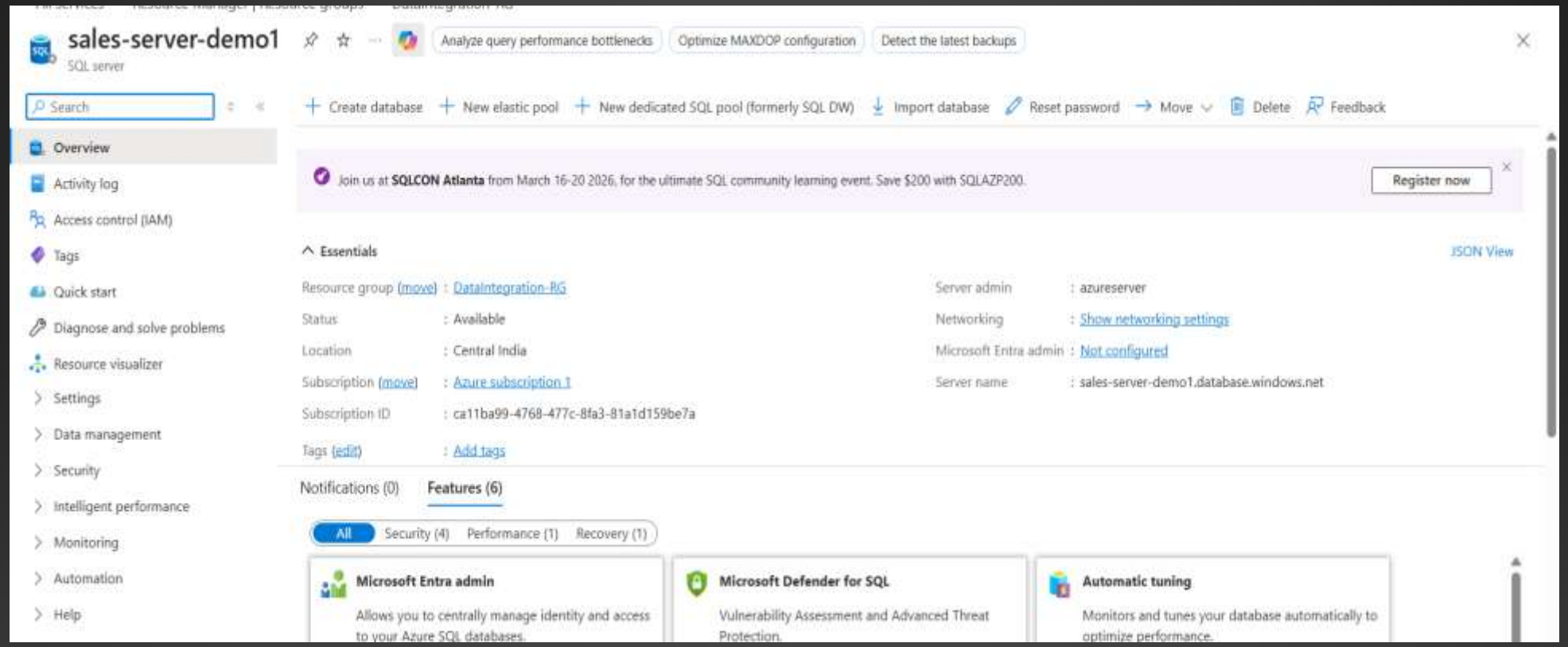
RAW DATA – CSV – UPLOAD A CSV FILE



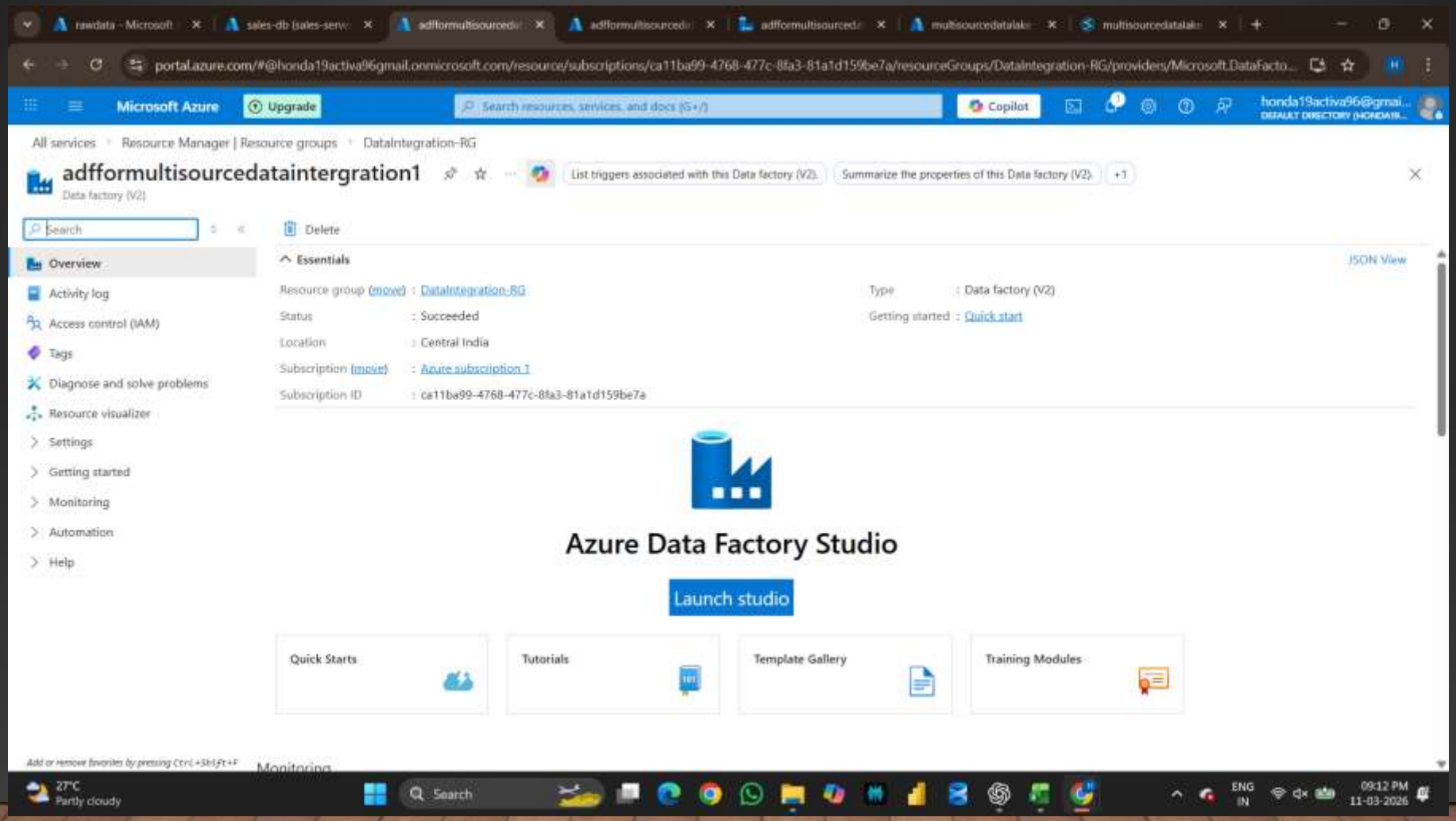
CREATE A SQL DATABASE



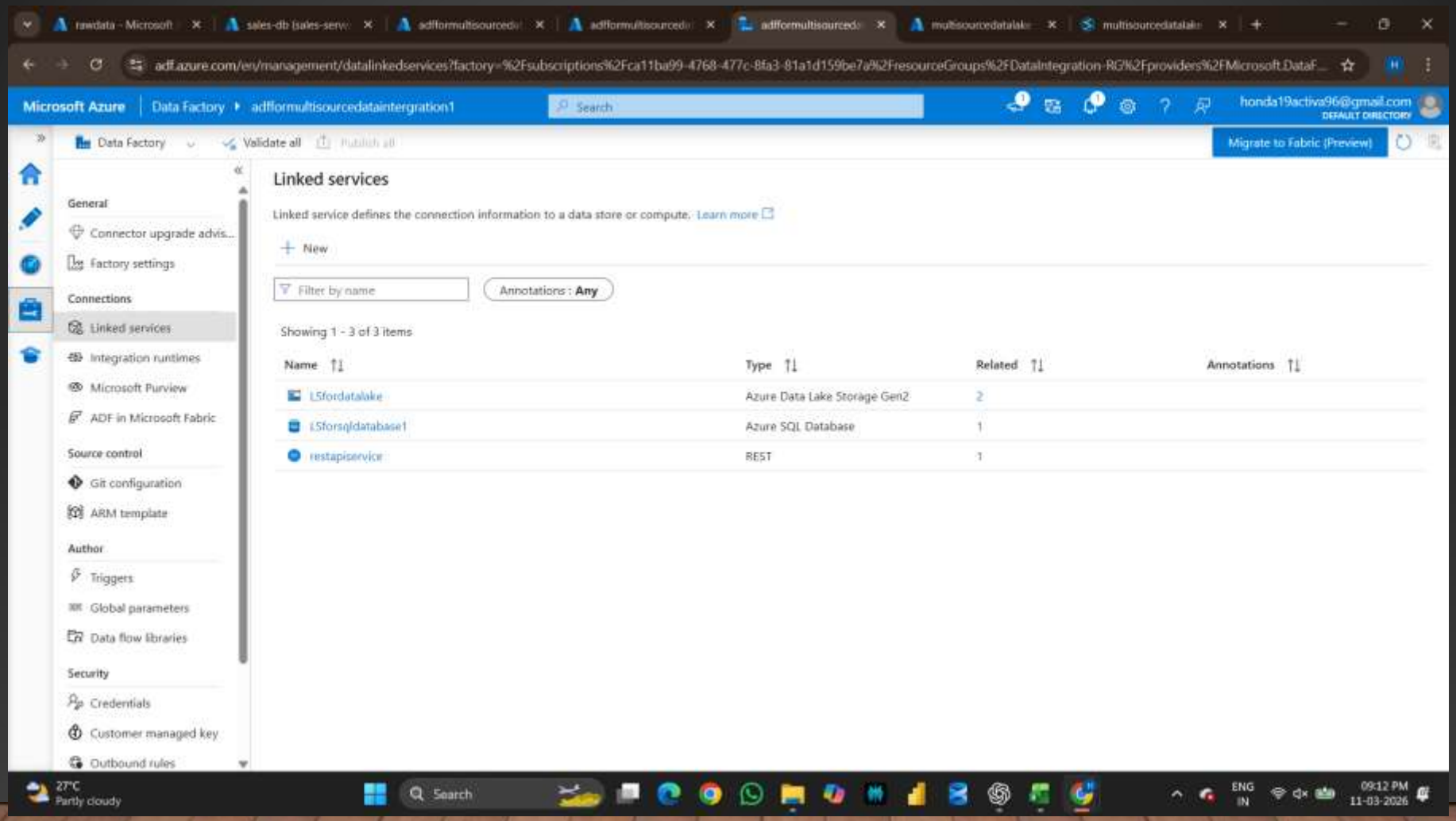
CREATE A SQL SERVER



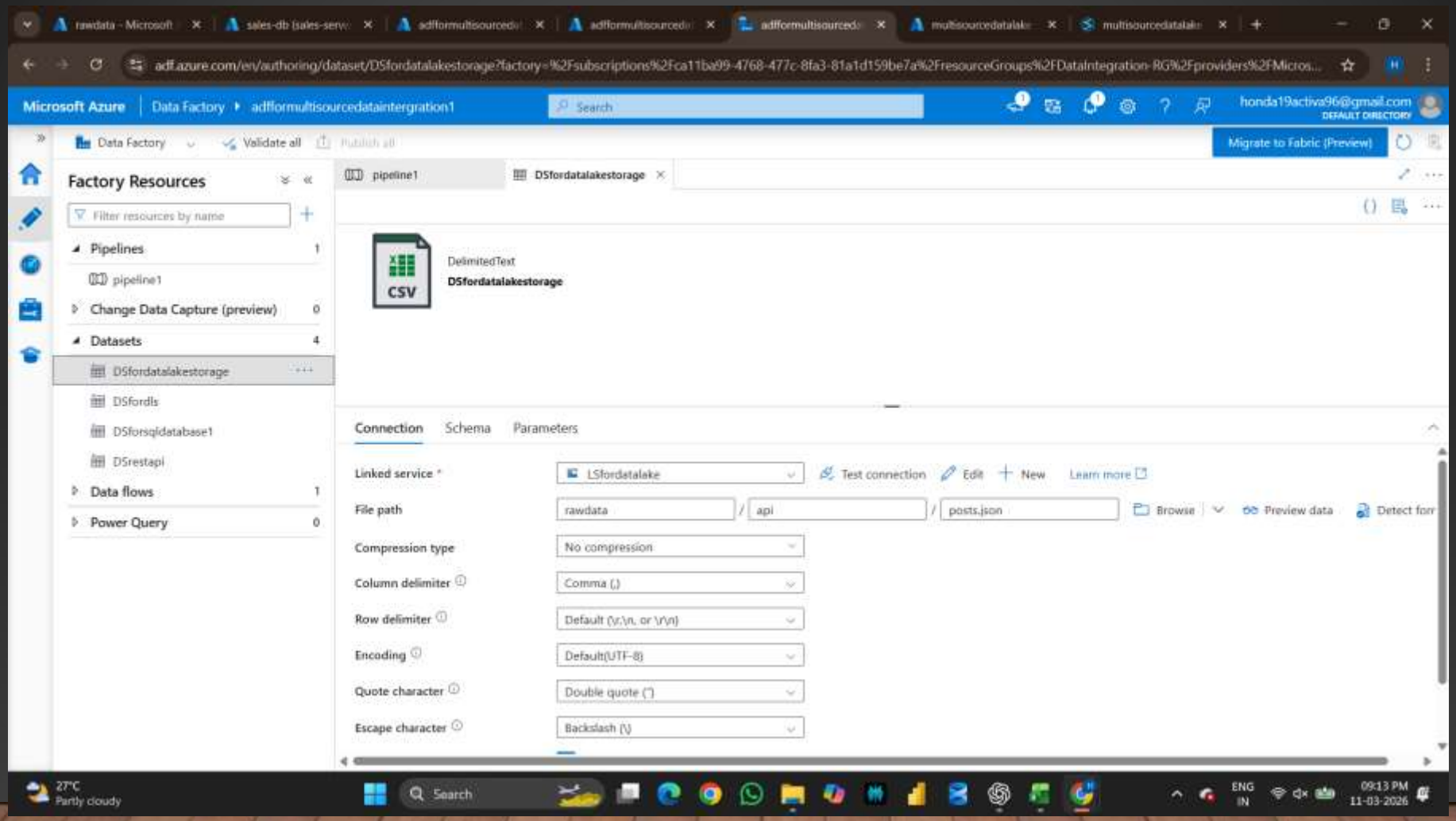
CREATE A AZURE DATA FACTORY



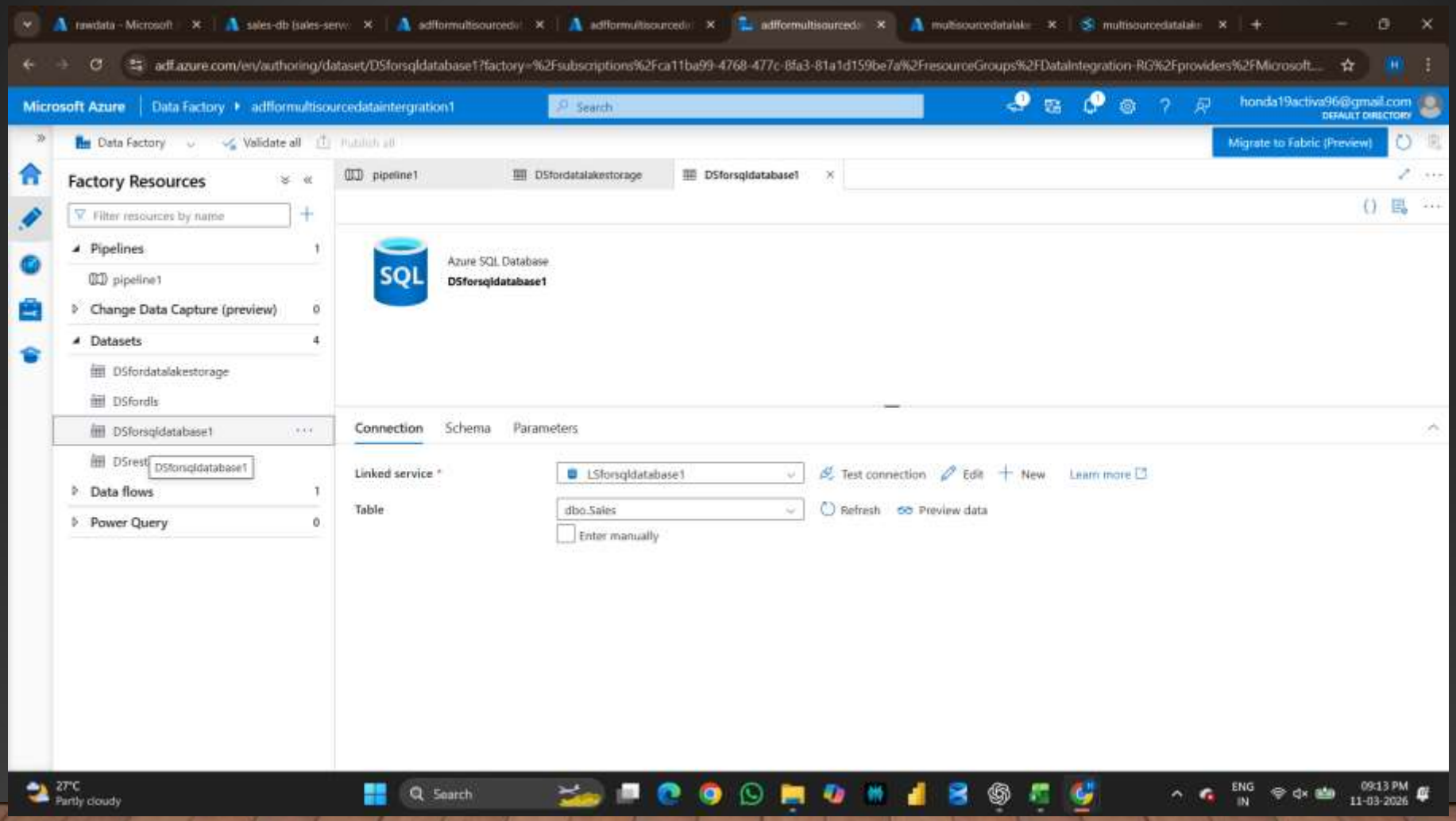
CREATE A LINKED SERVICES FOR DATALAKE,SQLDB,RESTAPI



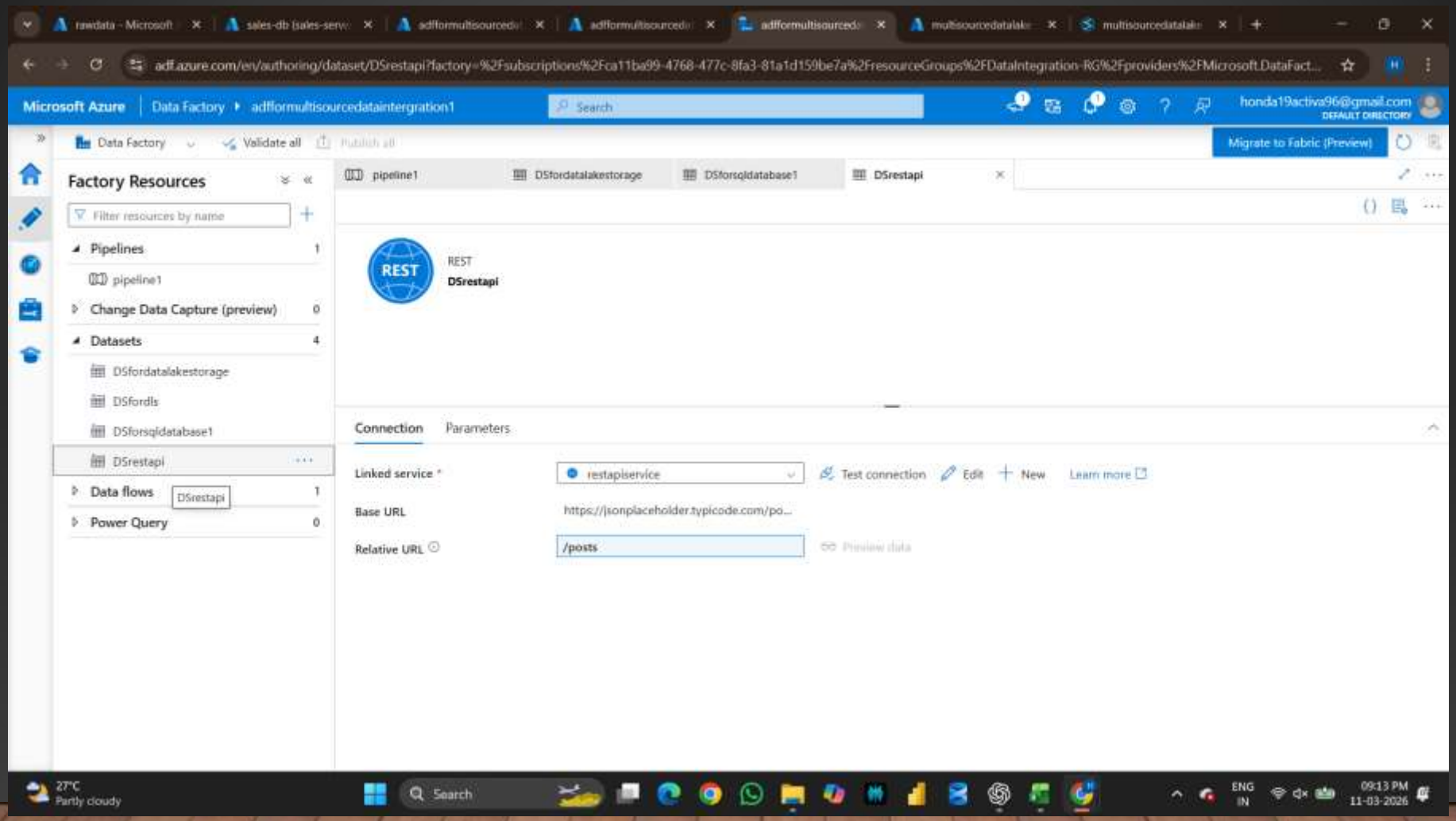
CREATE A DATASET FOR DATALAKE



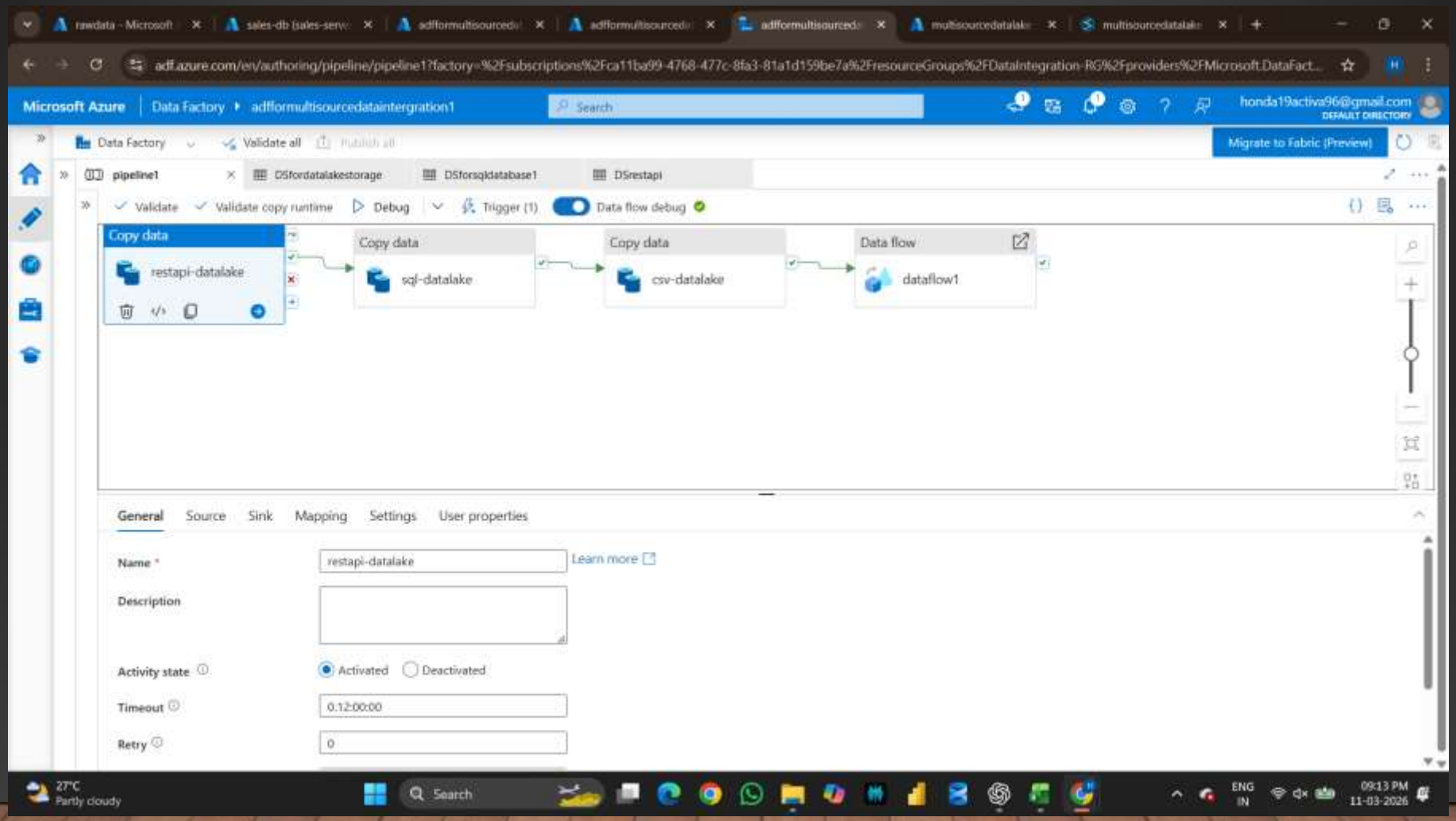
CREATE A DATASET FOR SQLDB



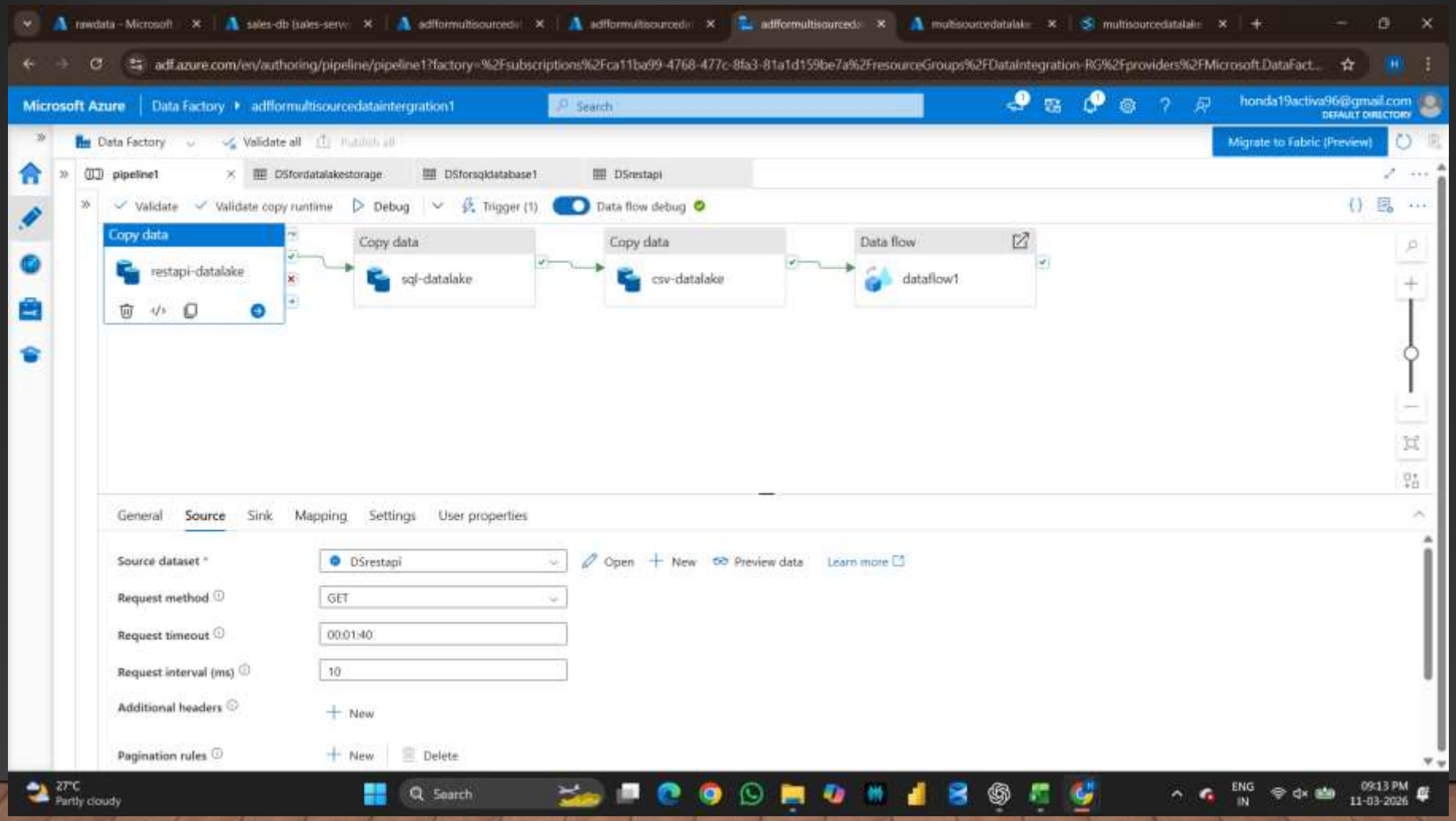
CREATE A DATASET FOR RESTAPI



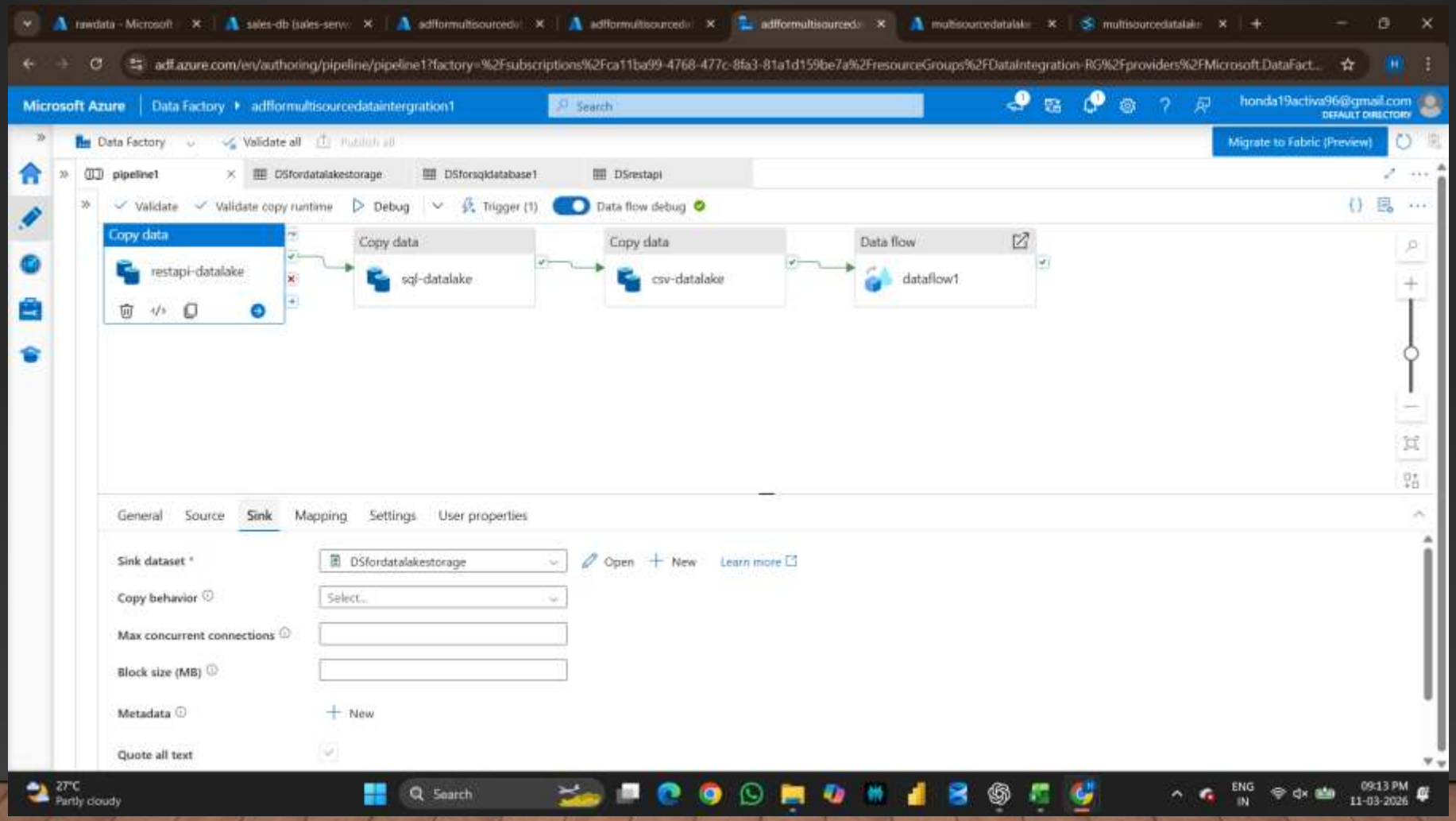
COPYDATA 1 –> RESTAPI - DATALAKE



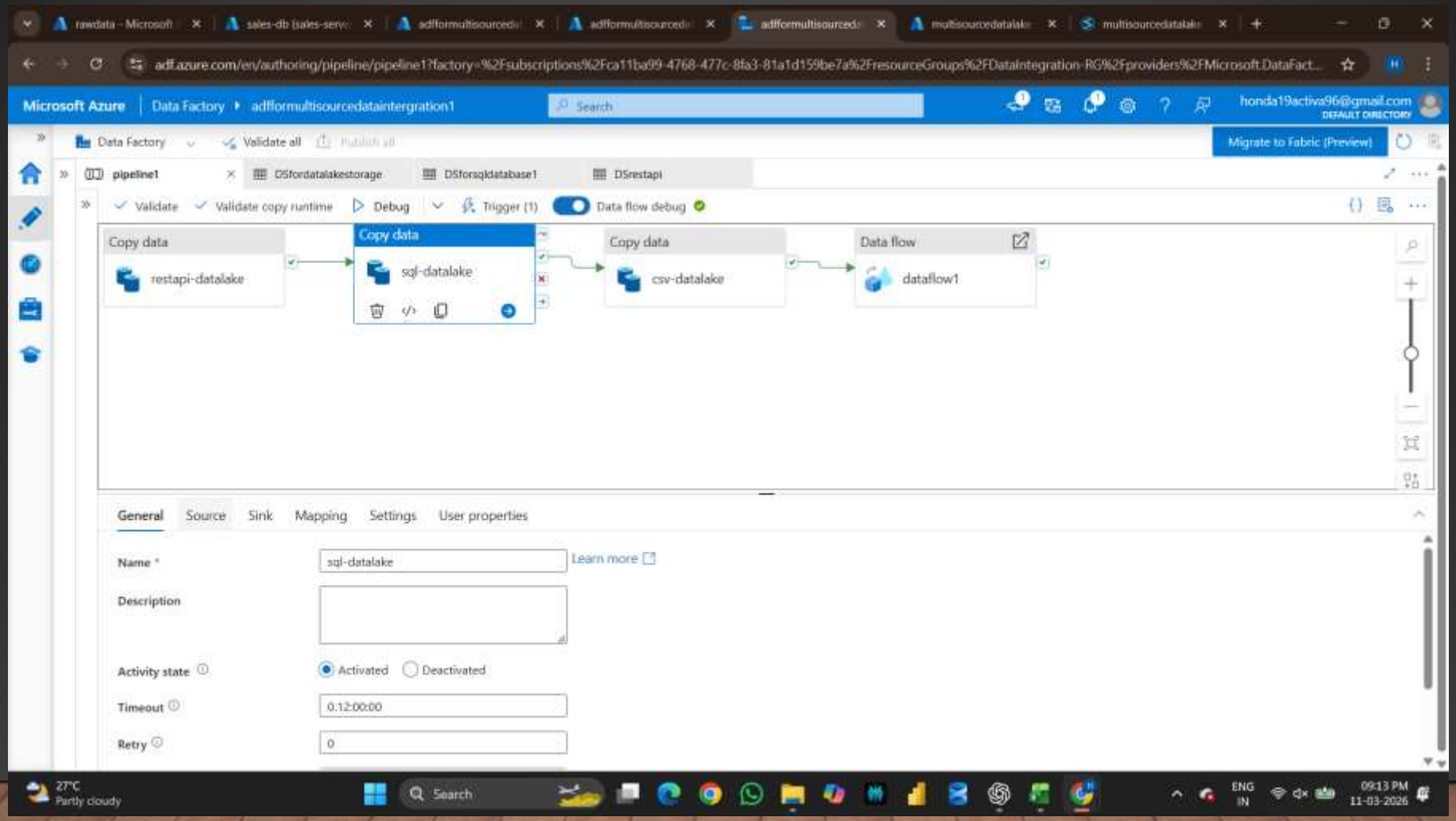
COPYDATA 1 –> RESTAPI - DATALAKE



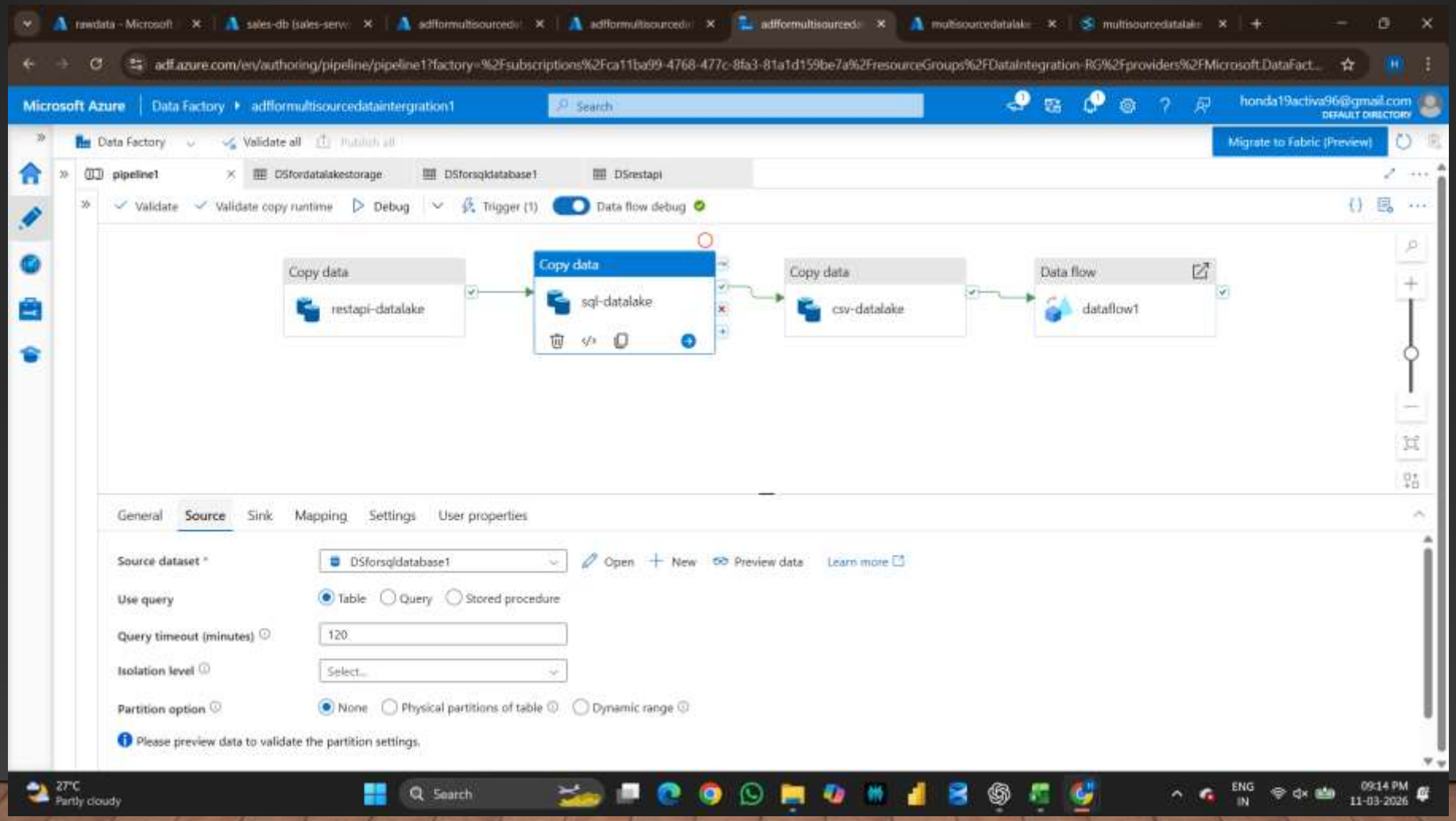
COPYDATA 1 –> RESTAPI - DATALAKE



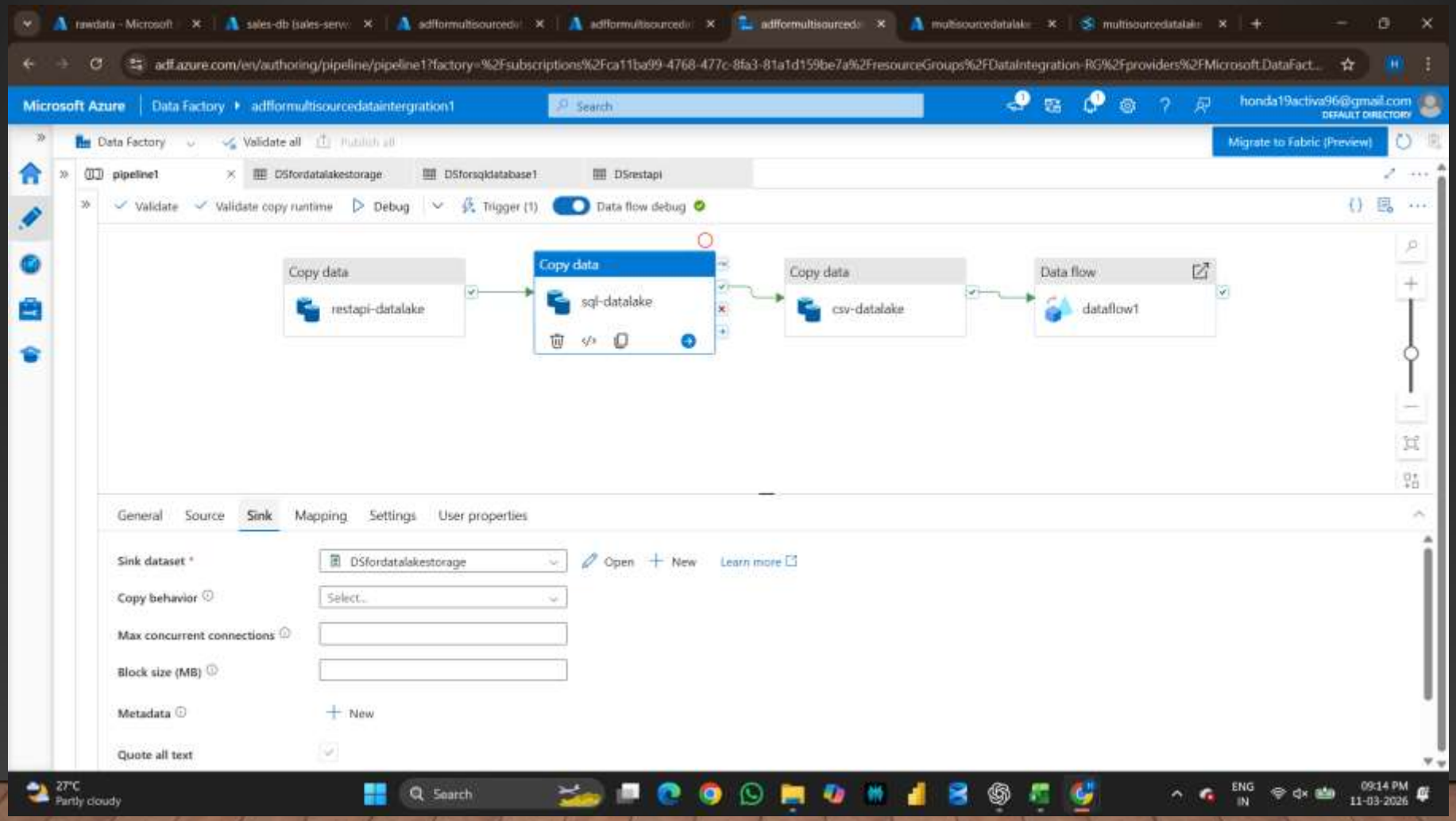
COPYDATA 2 –> SQLDB - DATALAKE



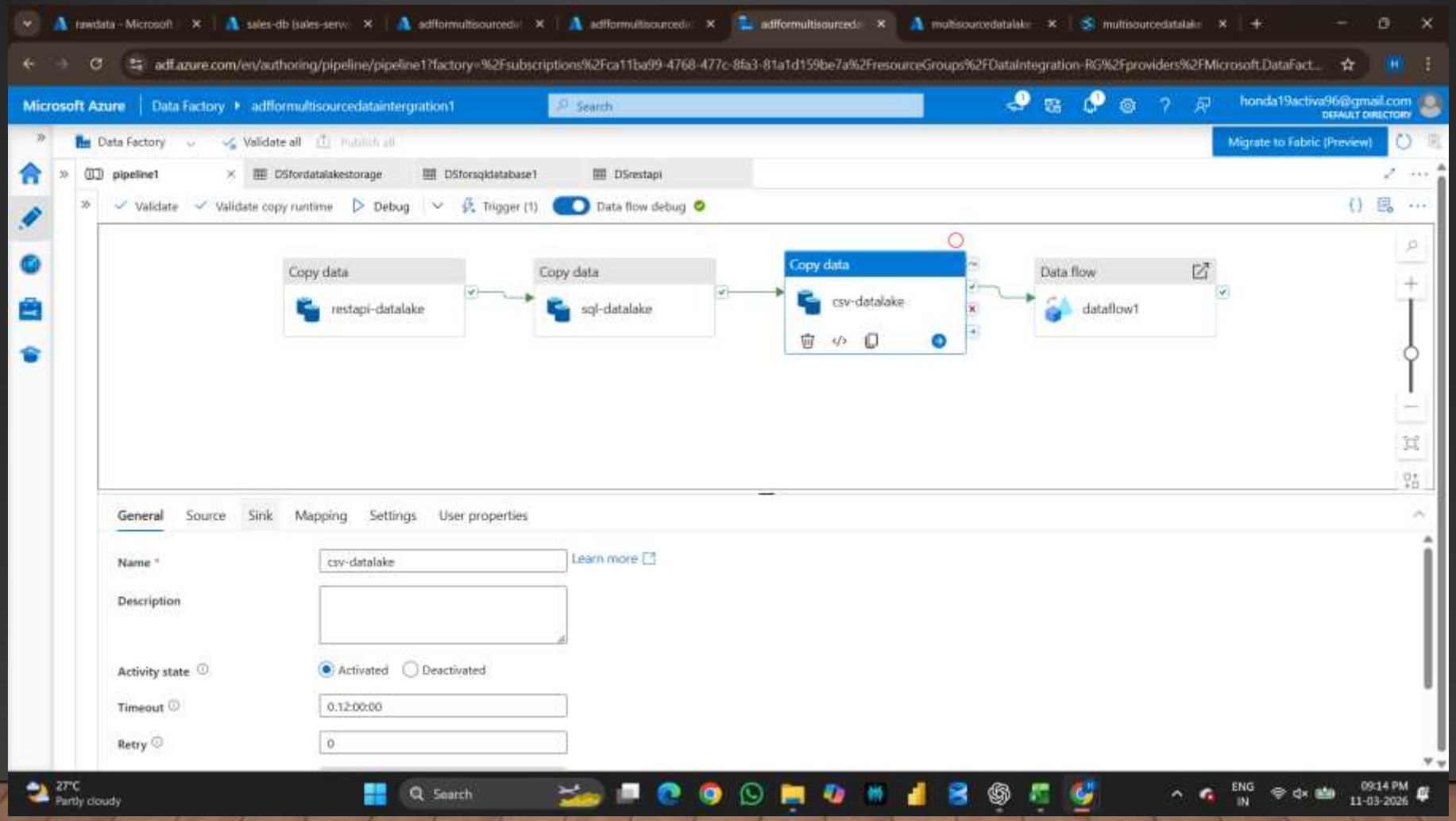
COPYDATA 2 –> SQLDB - DATALAKE



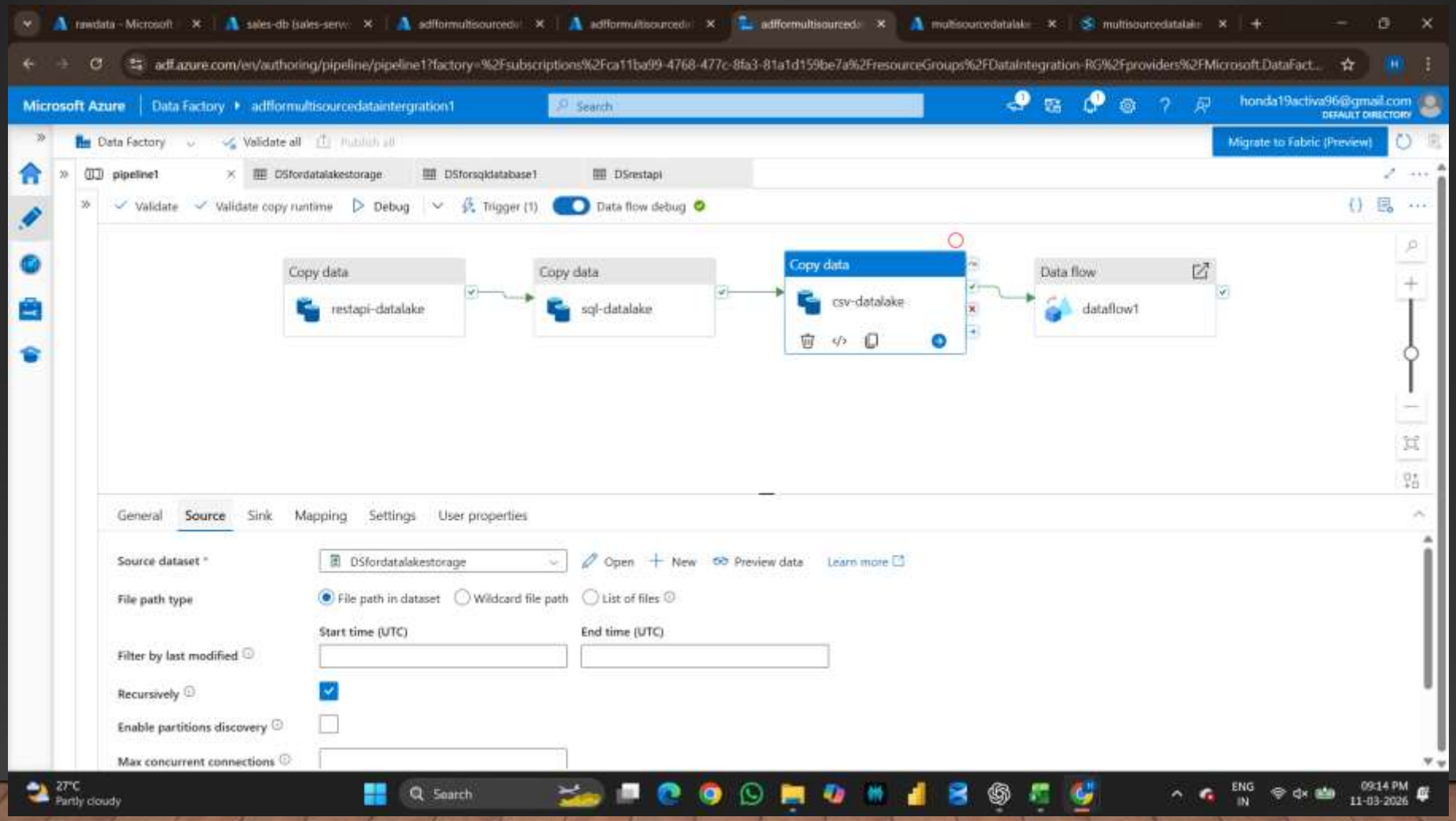
COPYDATA 2 –> SQLDB - DATALAKE



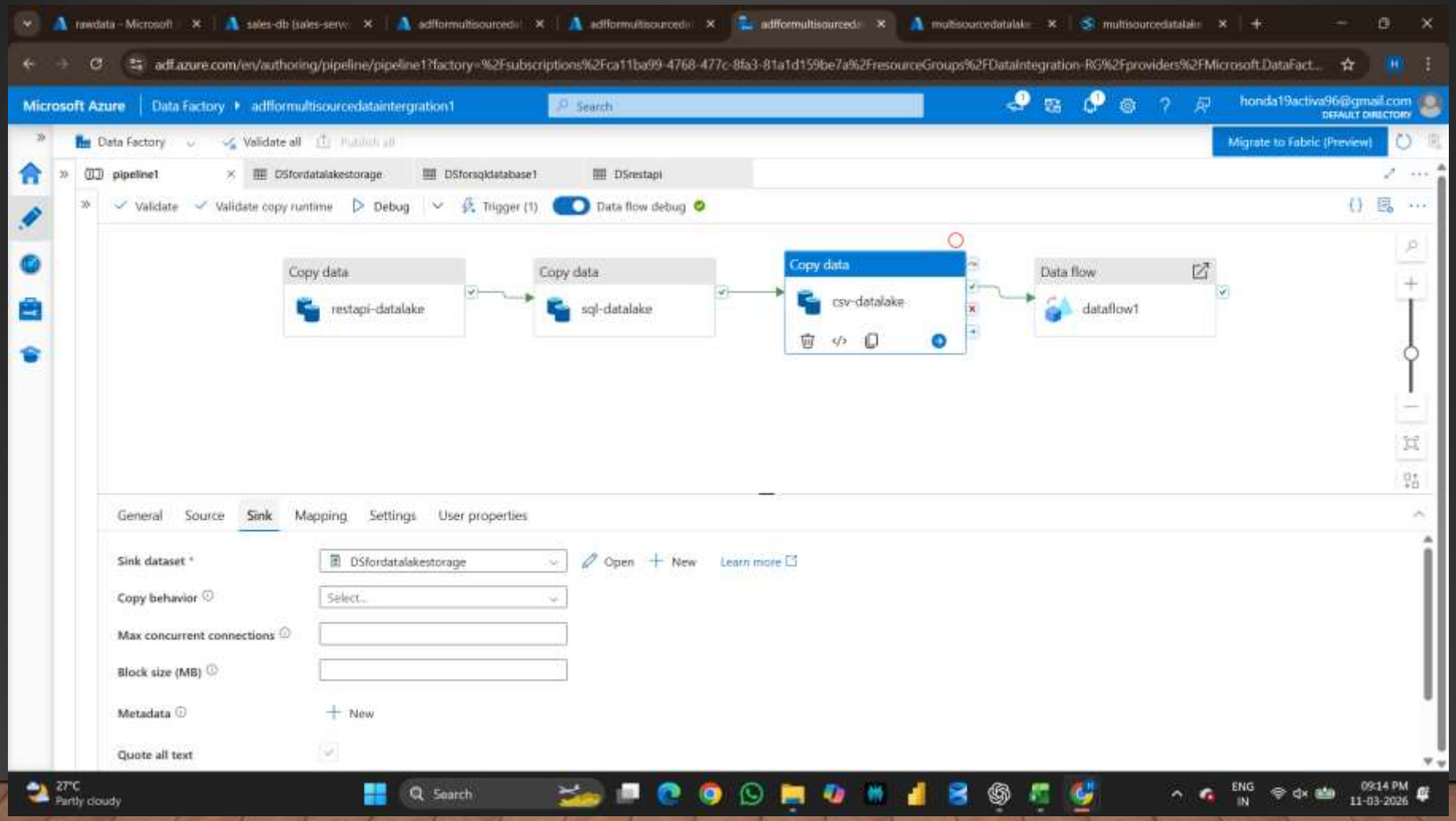
COPYDATA 3 –> CSV - DATALAKE



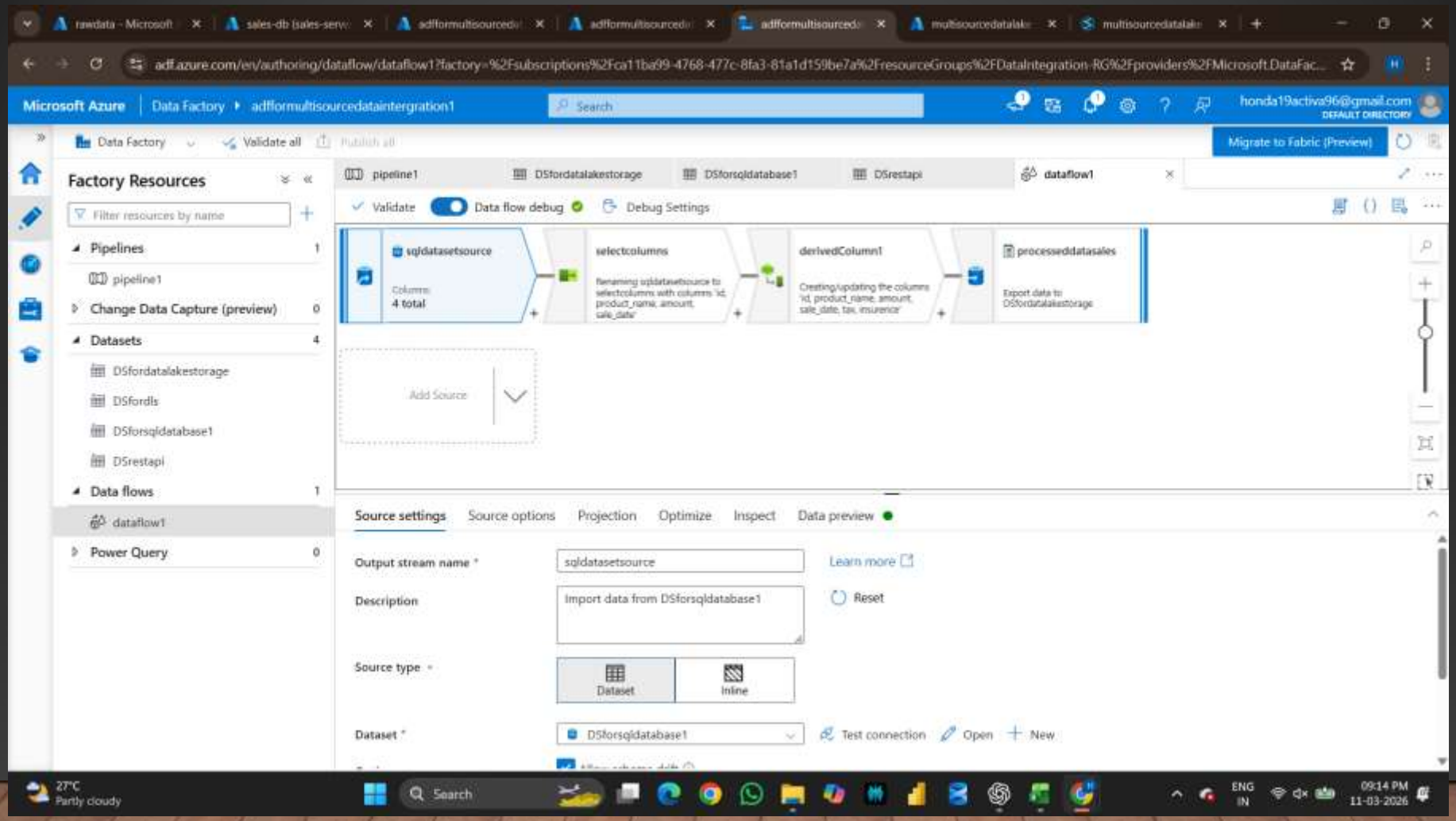
COPYDATA 3 –> CSV - DATALAKE



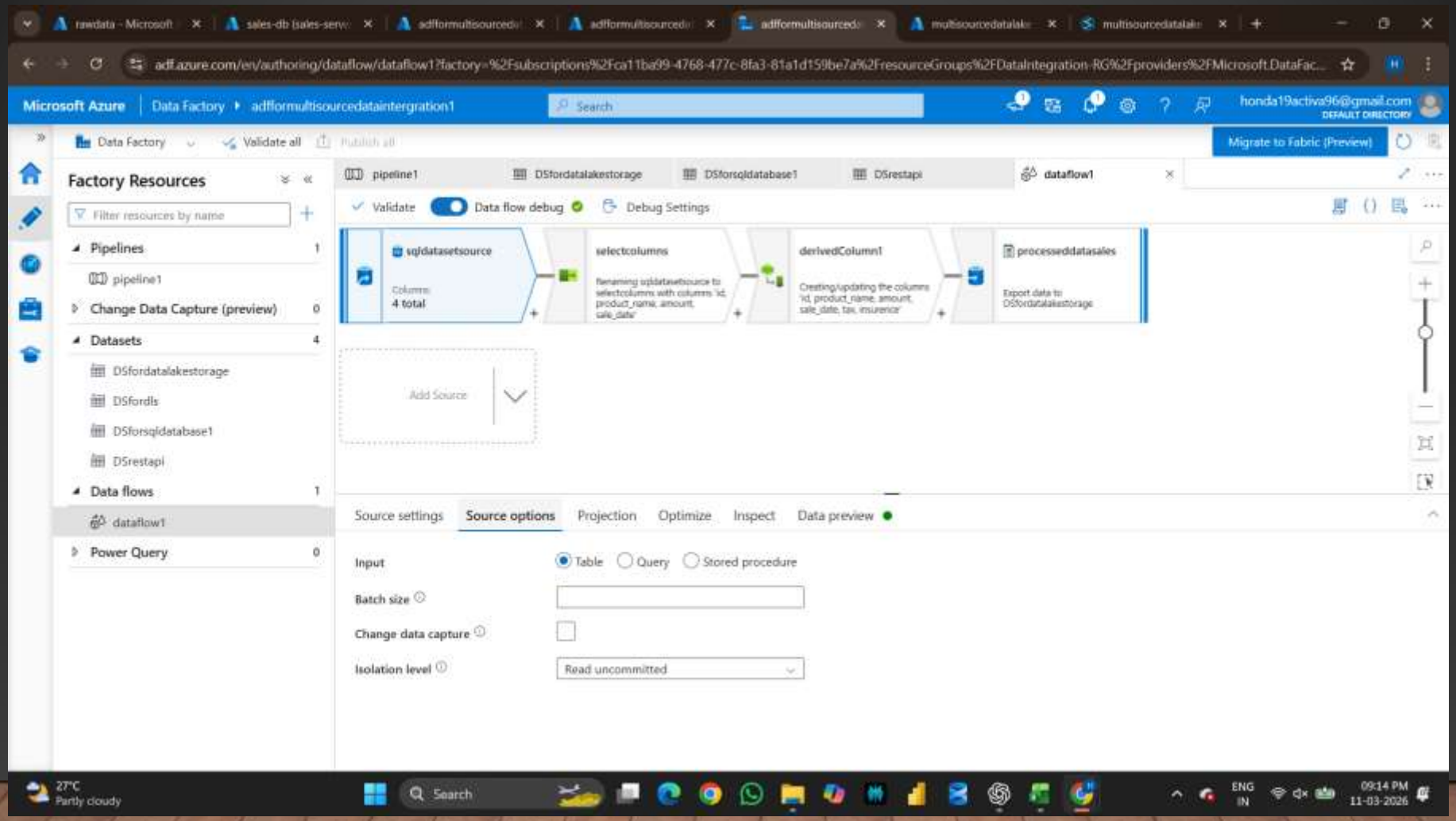
COPYDATA 3 –> CSV - DATALAKE



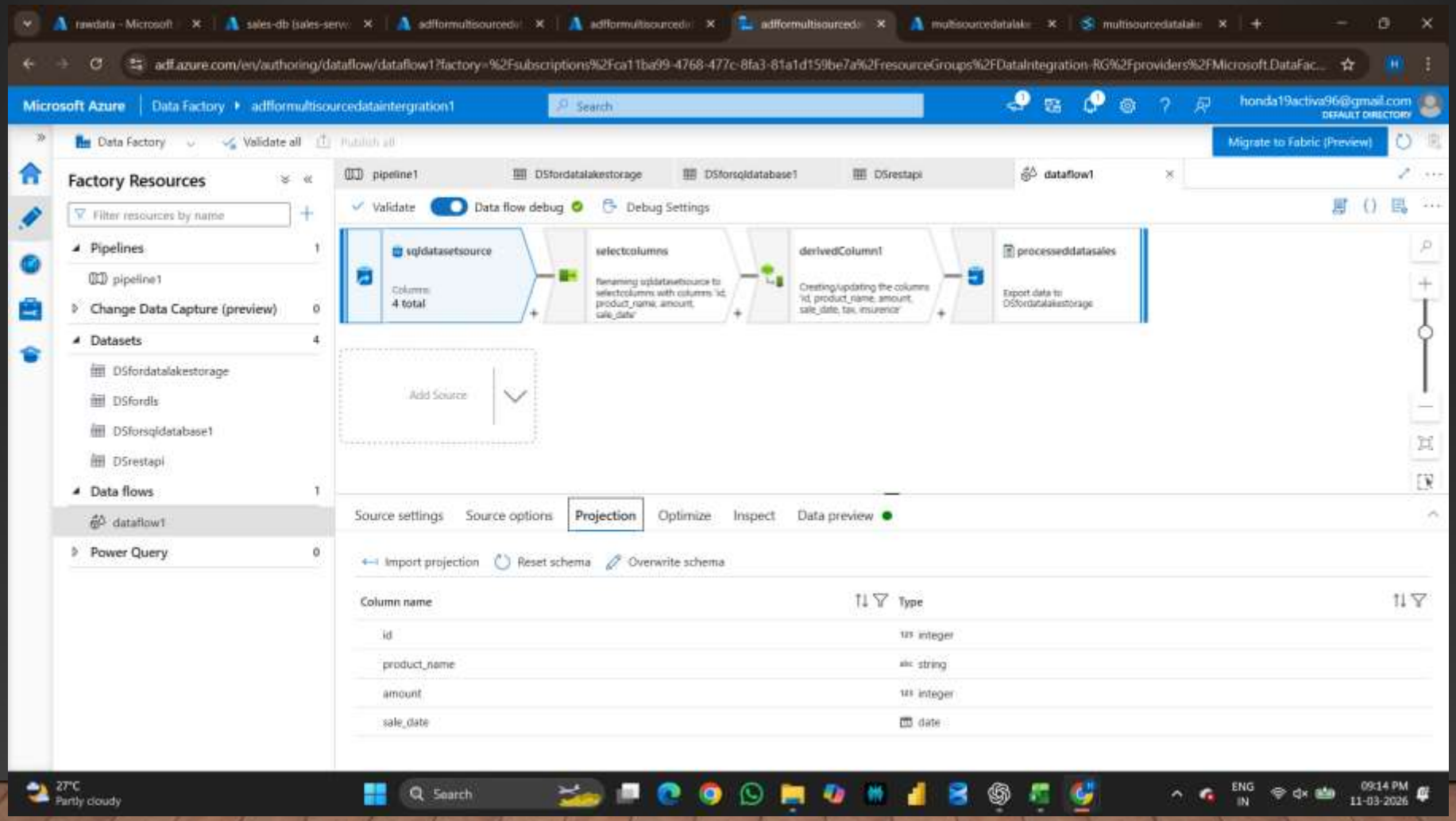
DATAFLOW 1 -> SQLDATABASE



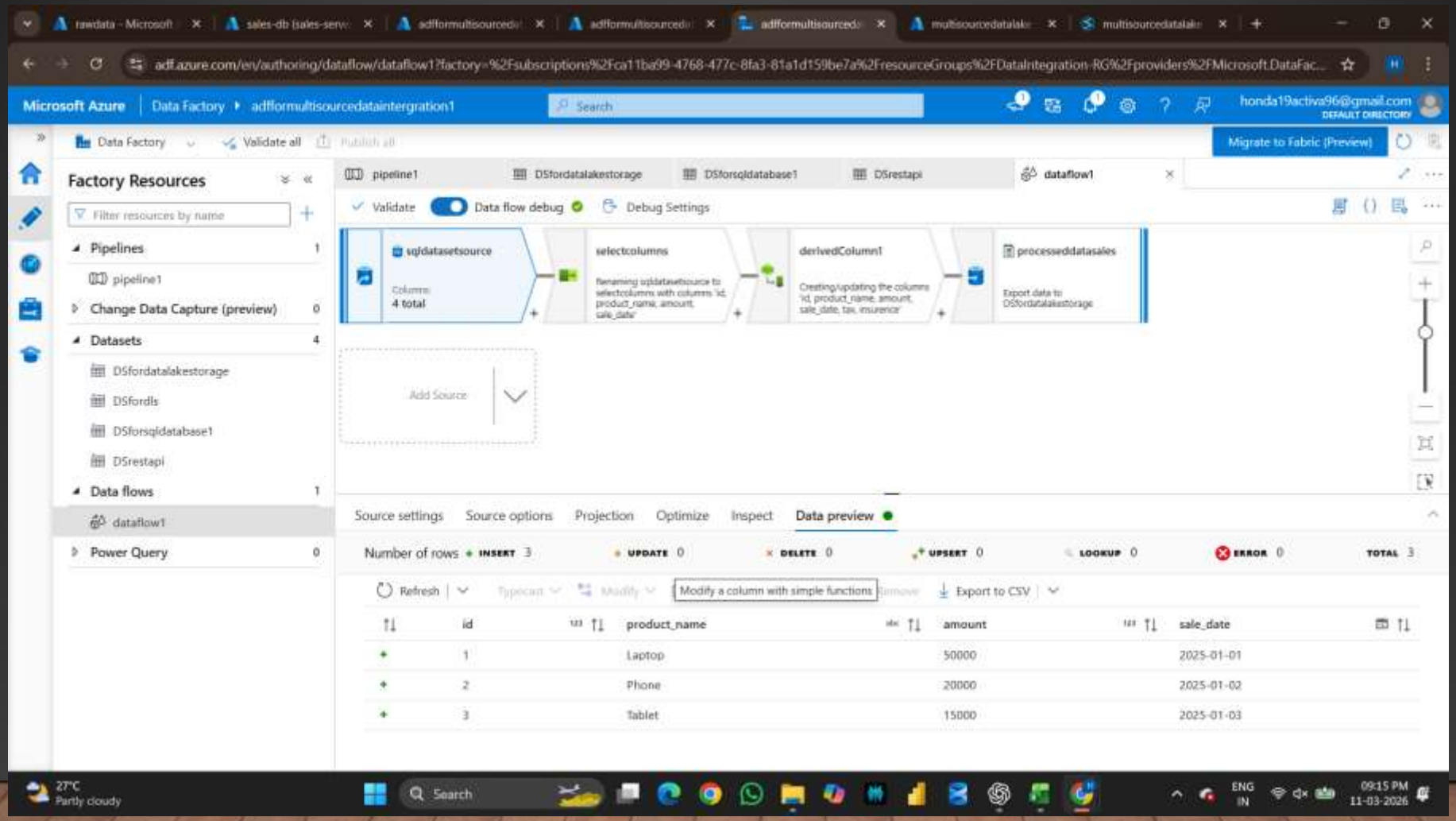
DATAFLOW 1 -> SQLDATABASE



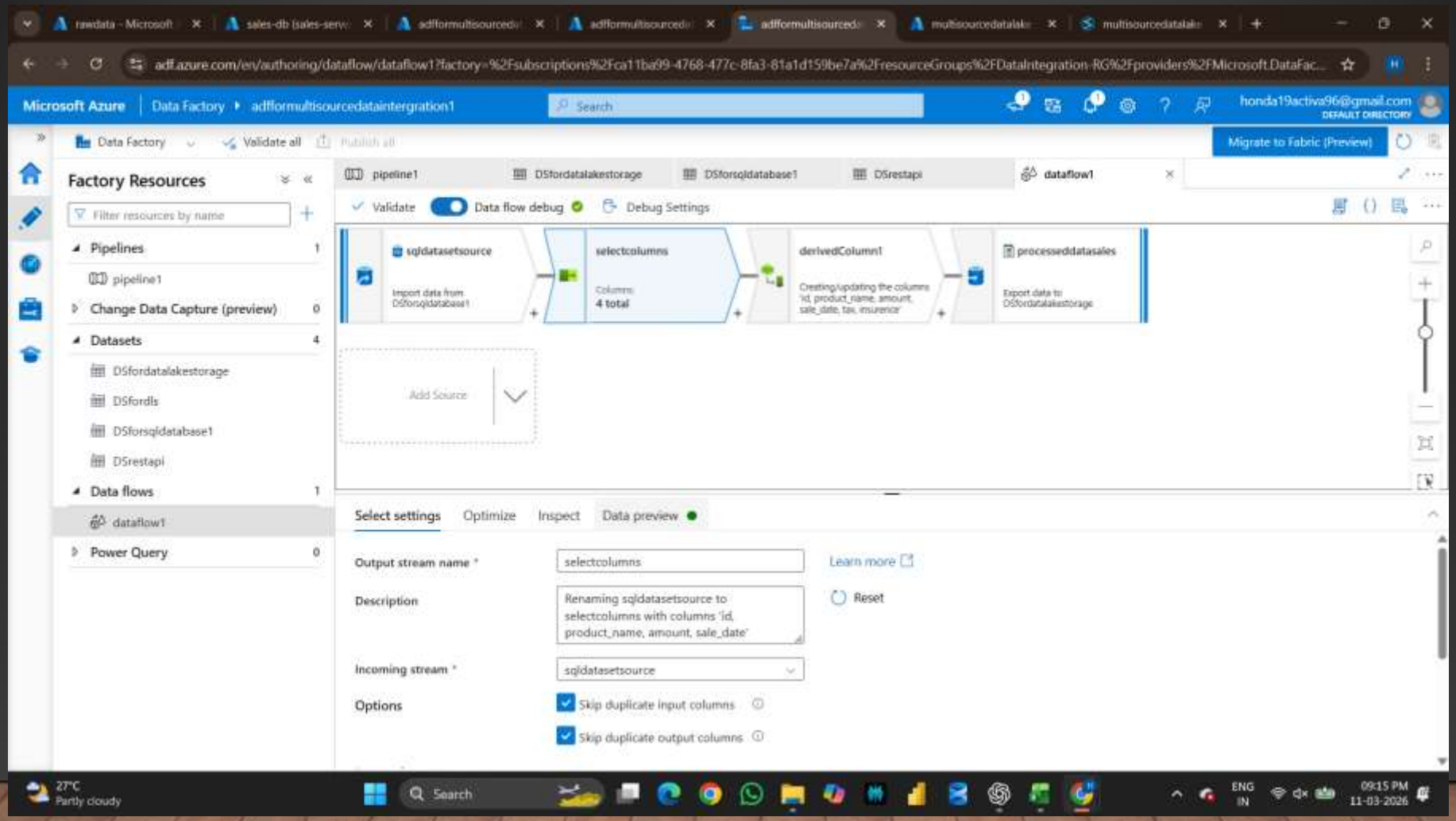
DATAFLOW 1 -> SOURCE (SQLDATABASE)



DATAFLOW 1 -> SQLDATABASE



DATAFLOW 1 -> SELECT COLUMNS



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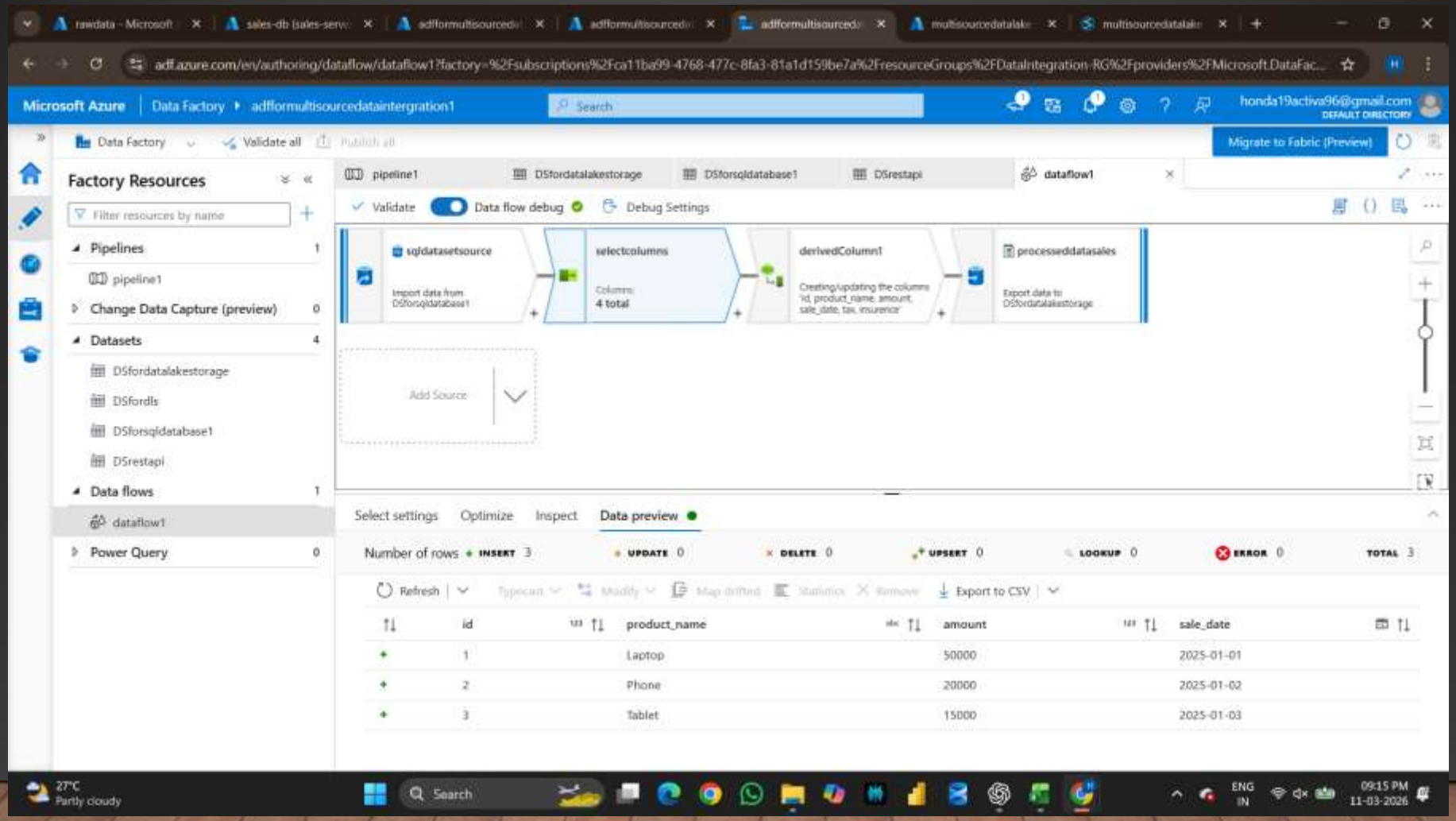
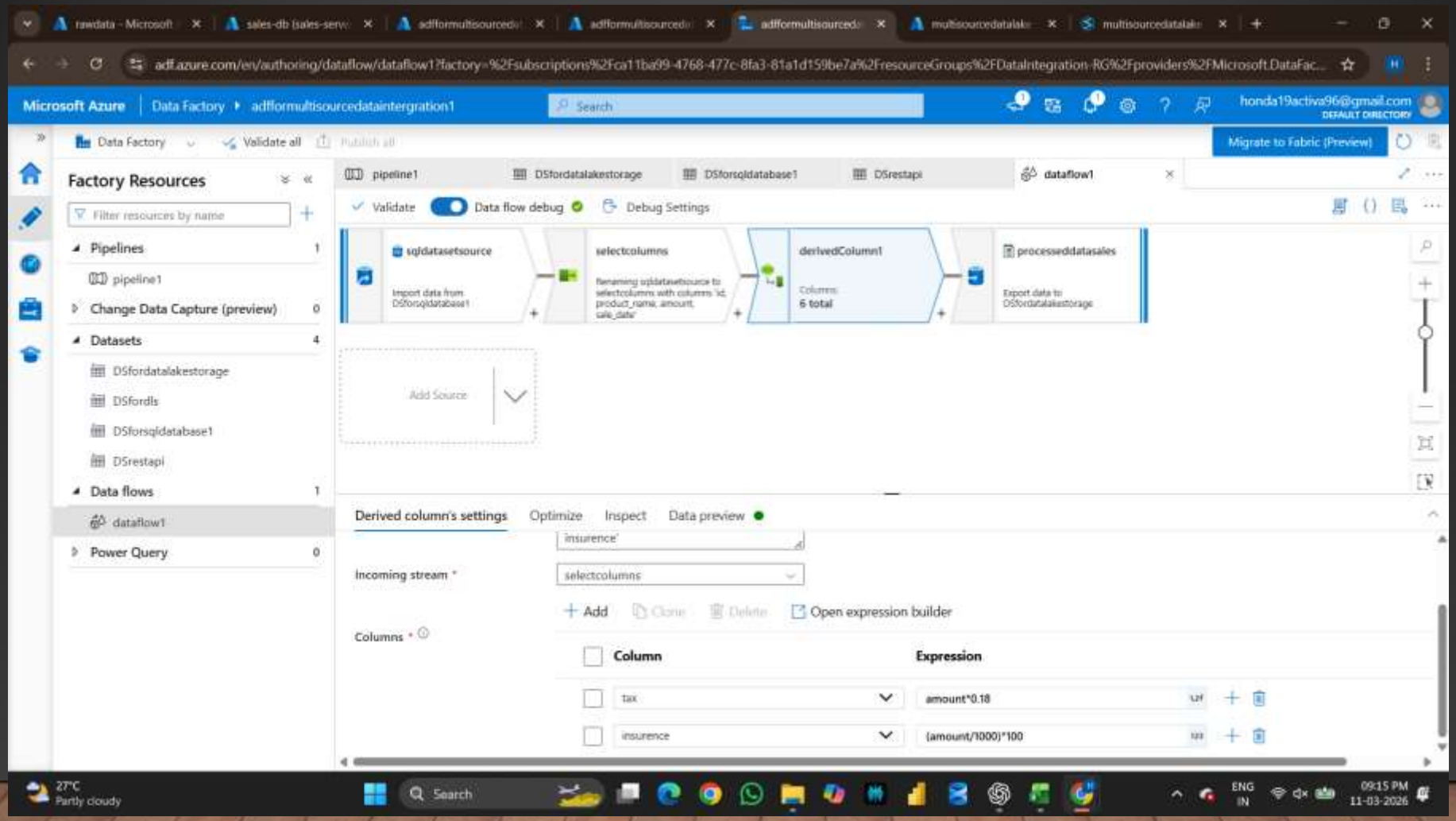
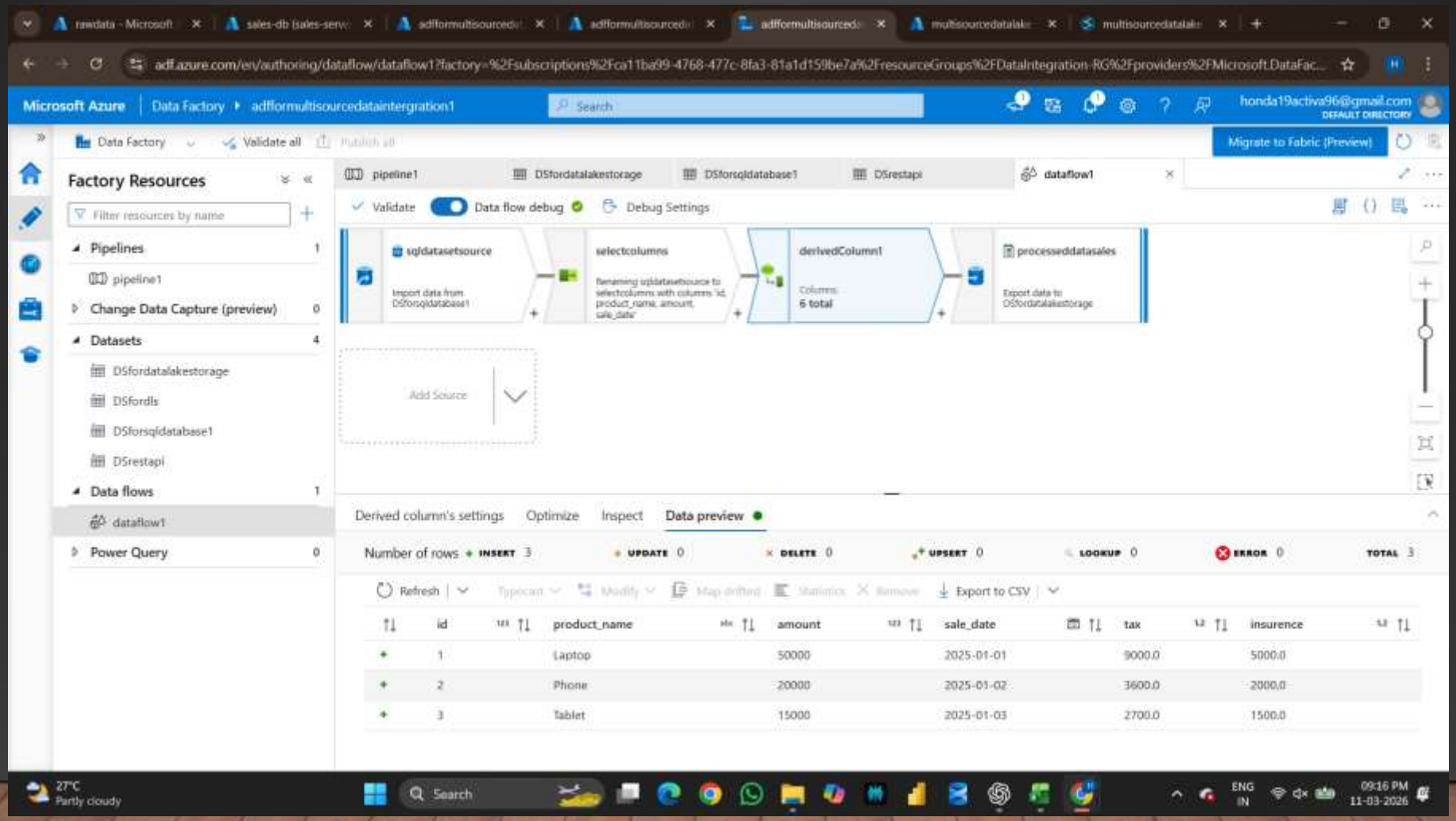


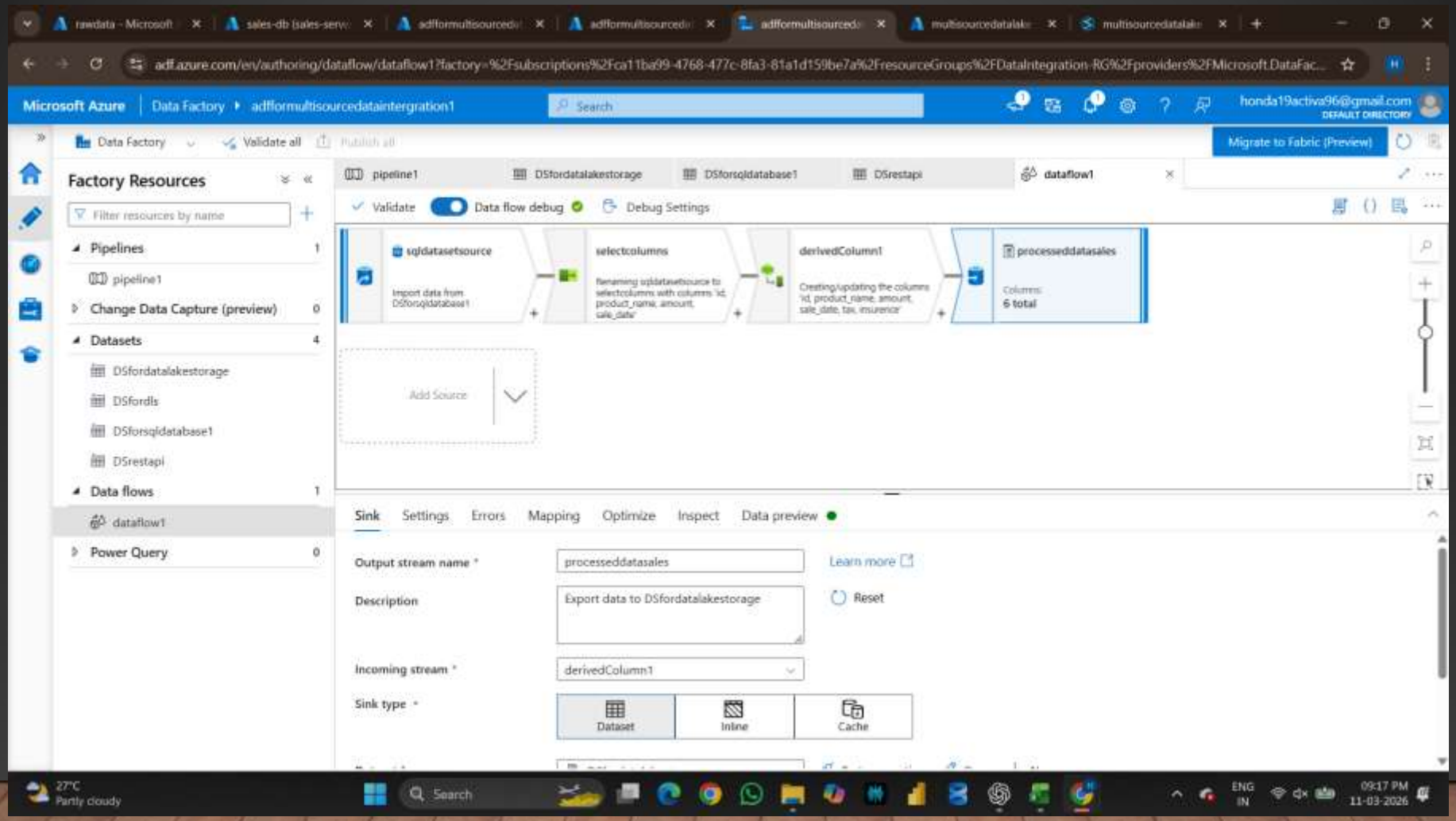
DATAFLOW 1 -> DERIEVED COLUMNS



DATAFLOW 1 -> DERIEVED COLUMNS



DATAFLOW 1 -> SINK (PROCESSED DATA SALES)



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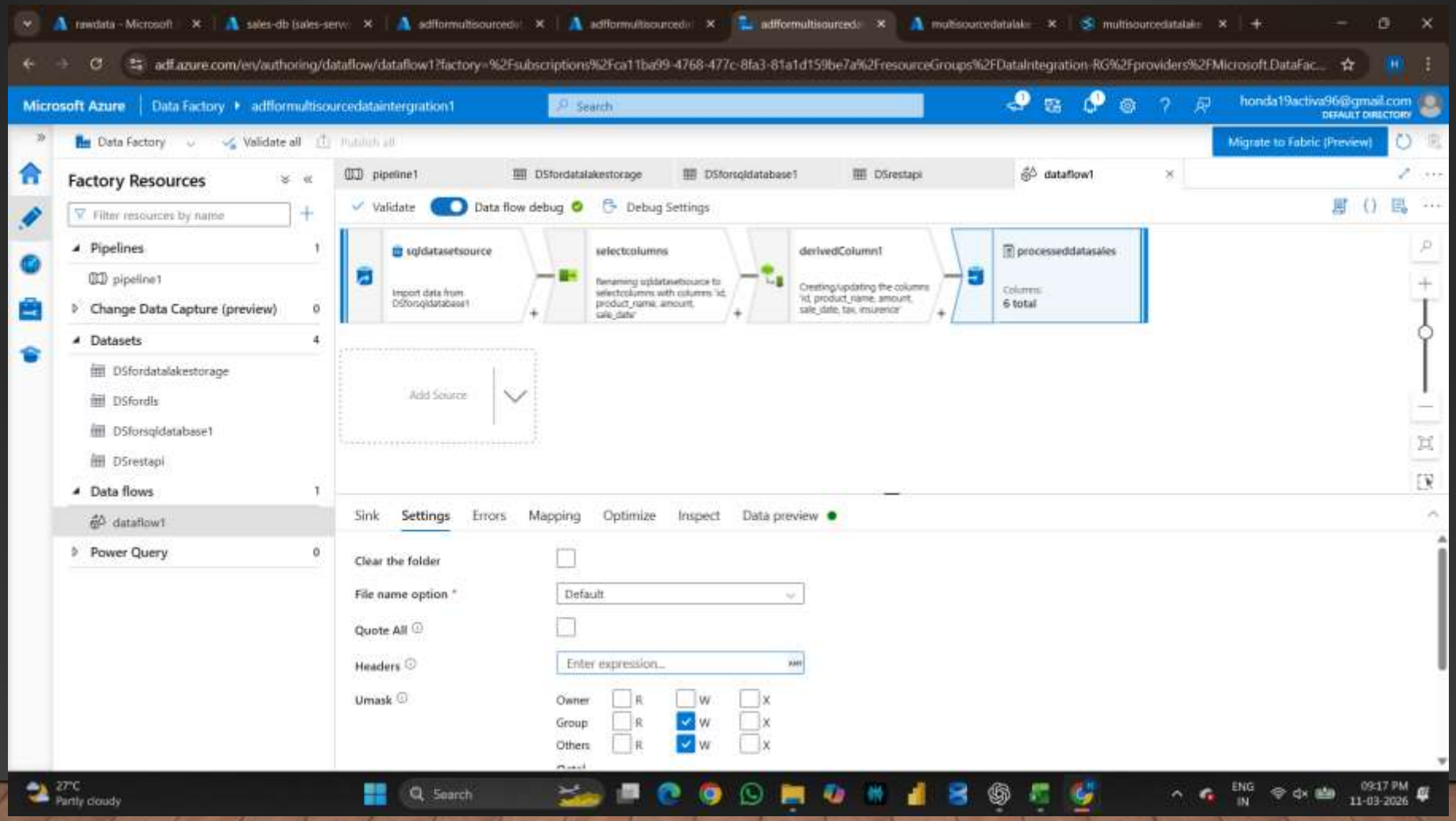

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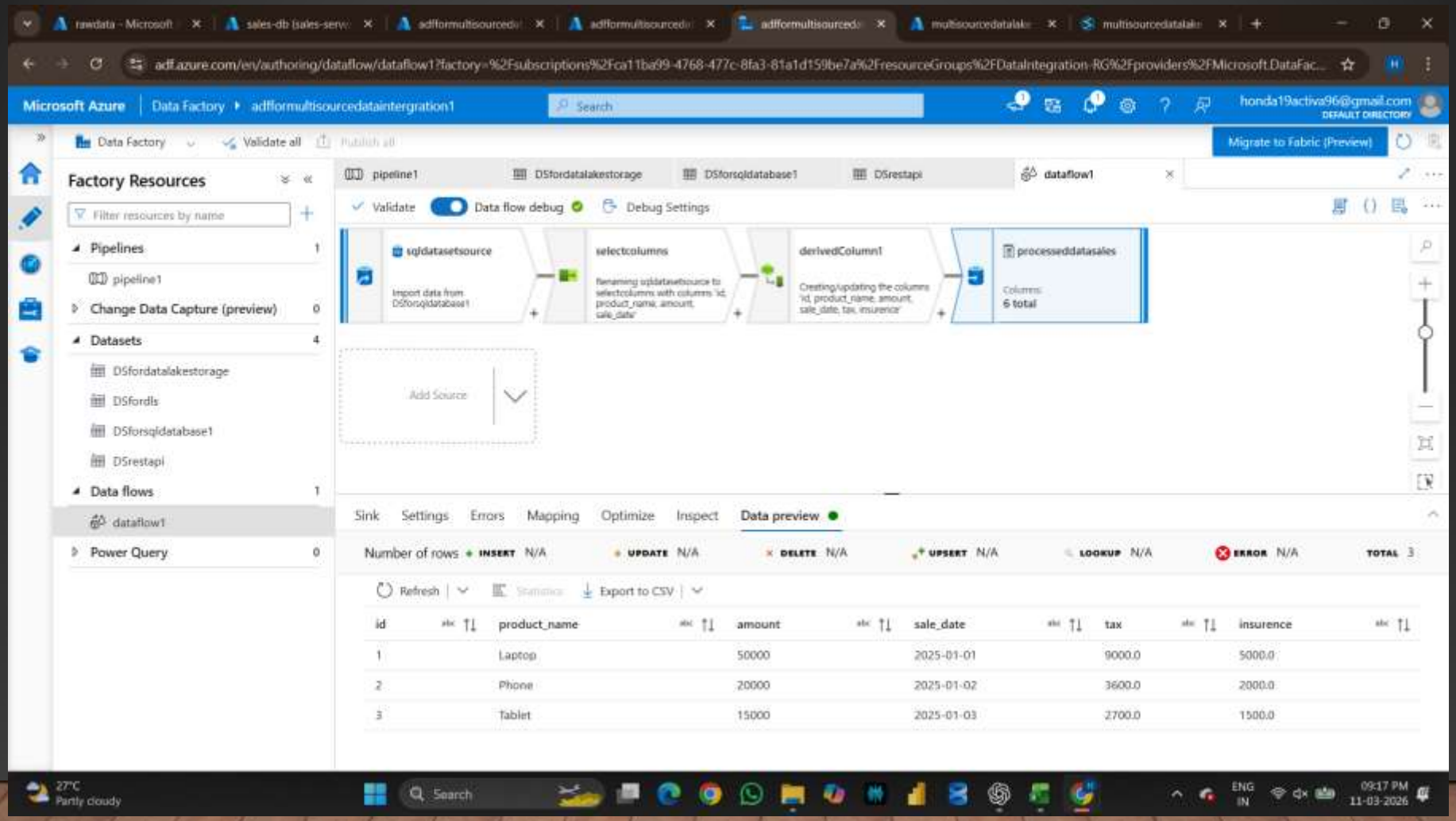
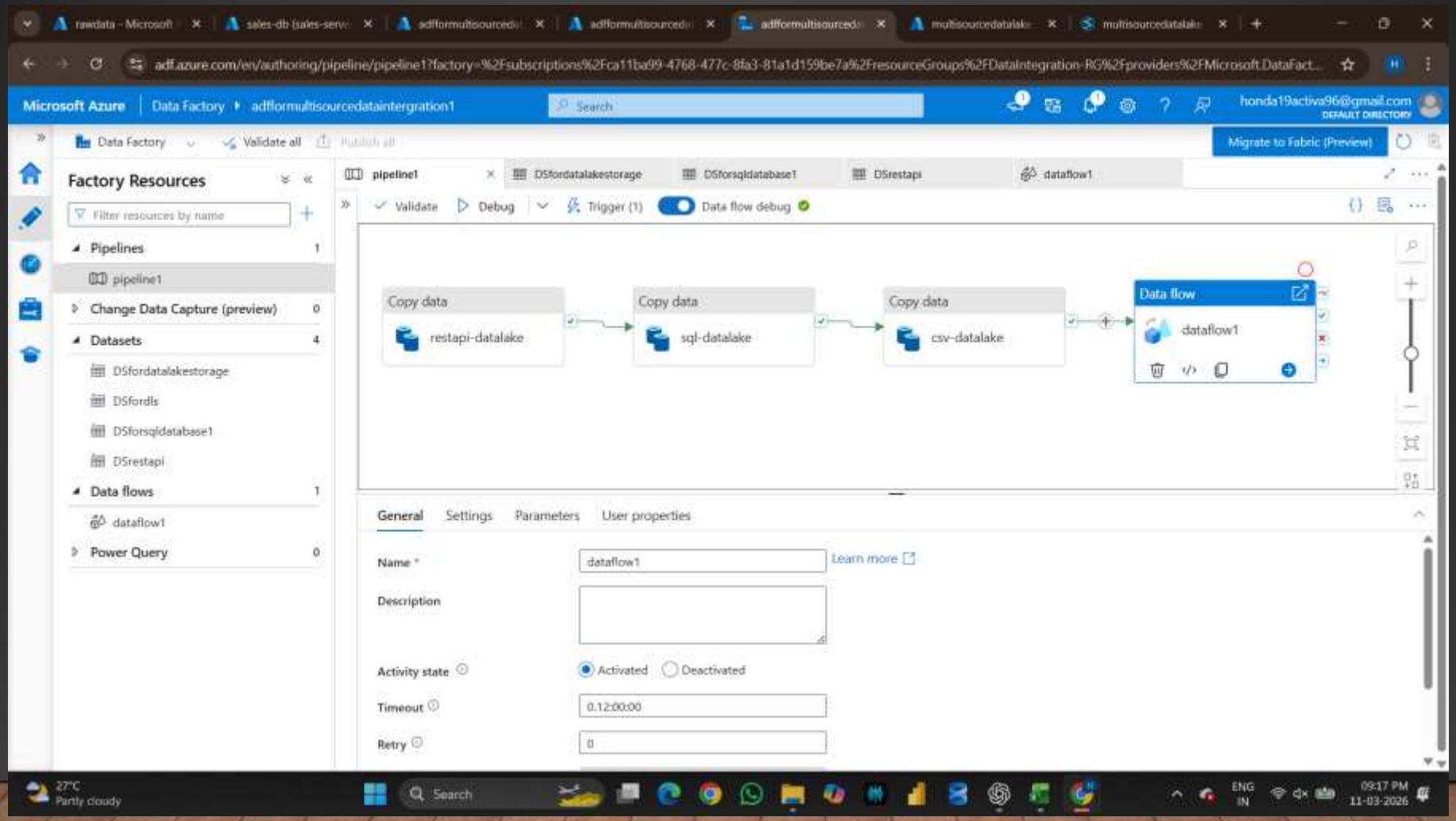


NOW YOU CAN CONNECT ALL THE COPY DATA ACTIVITY AND DATAFLOW



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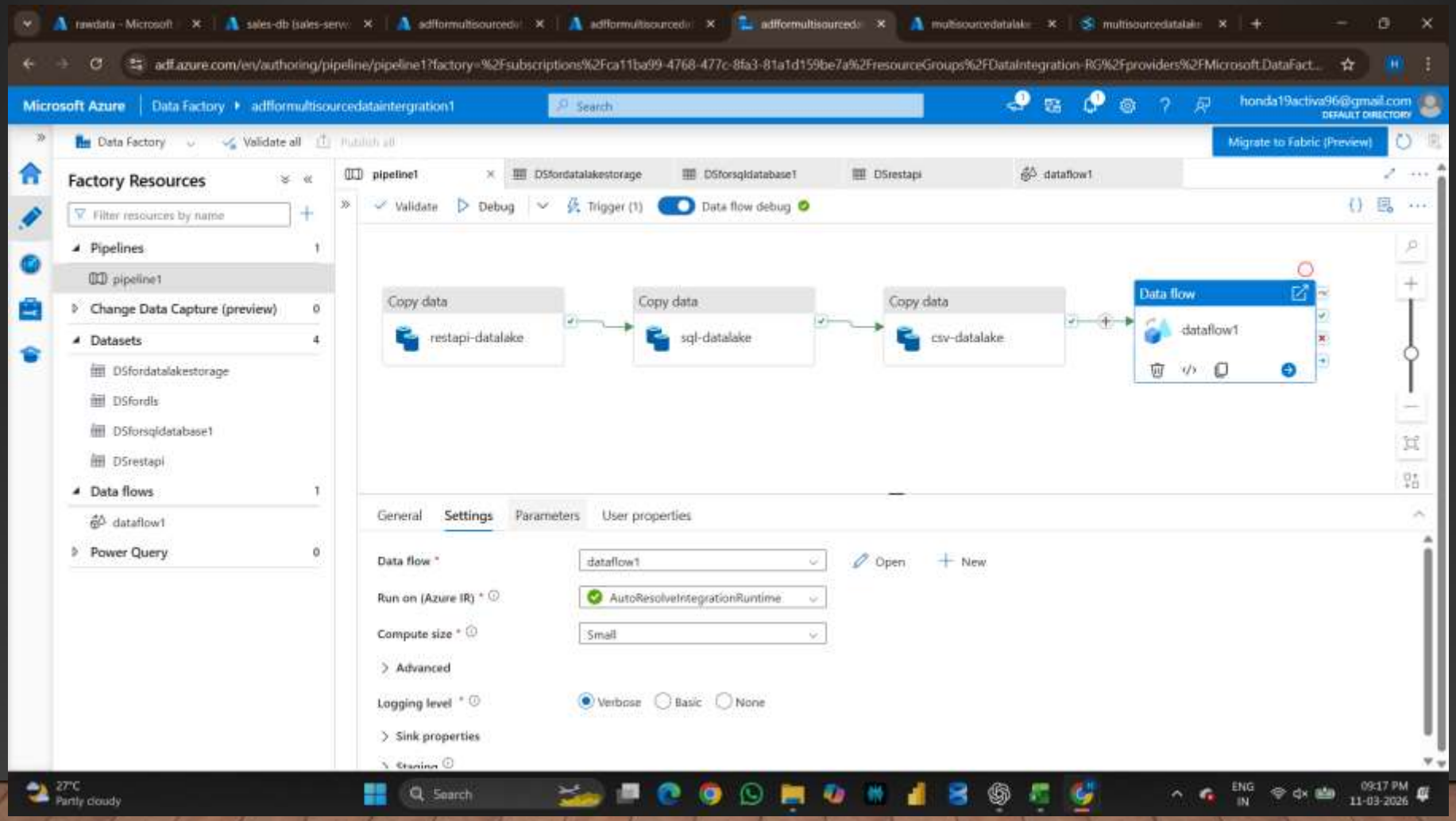
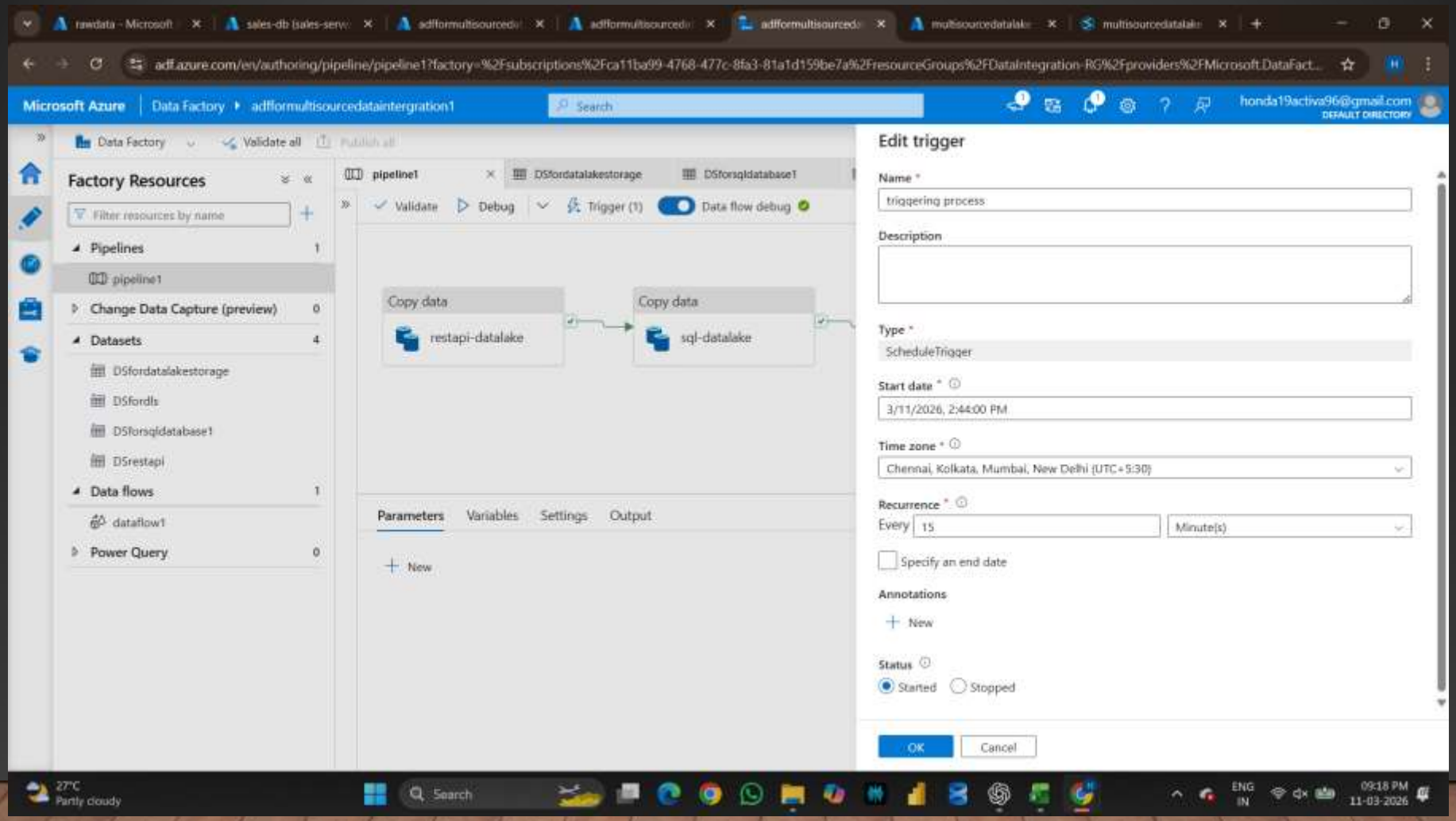
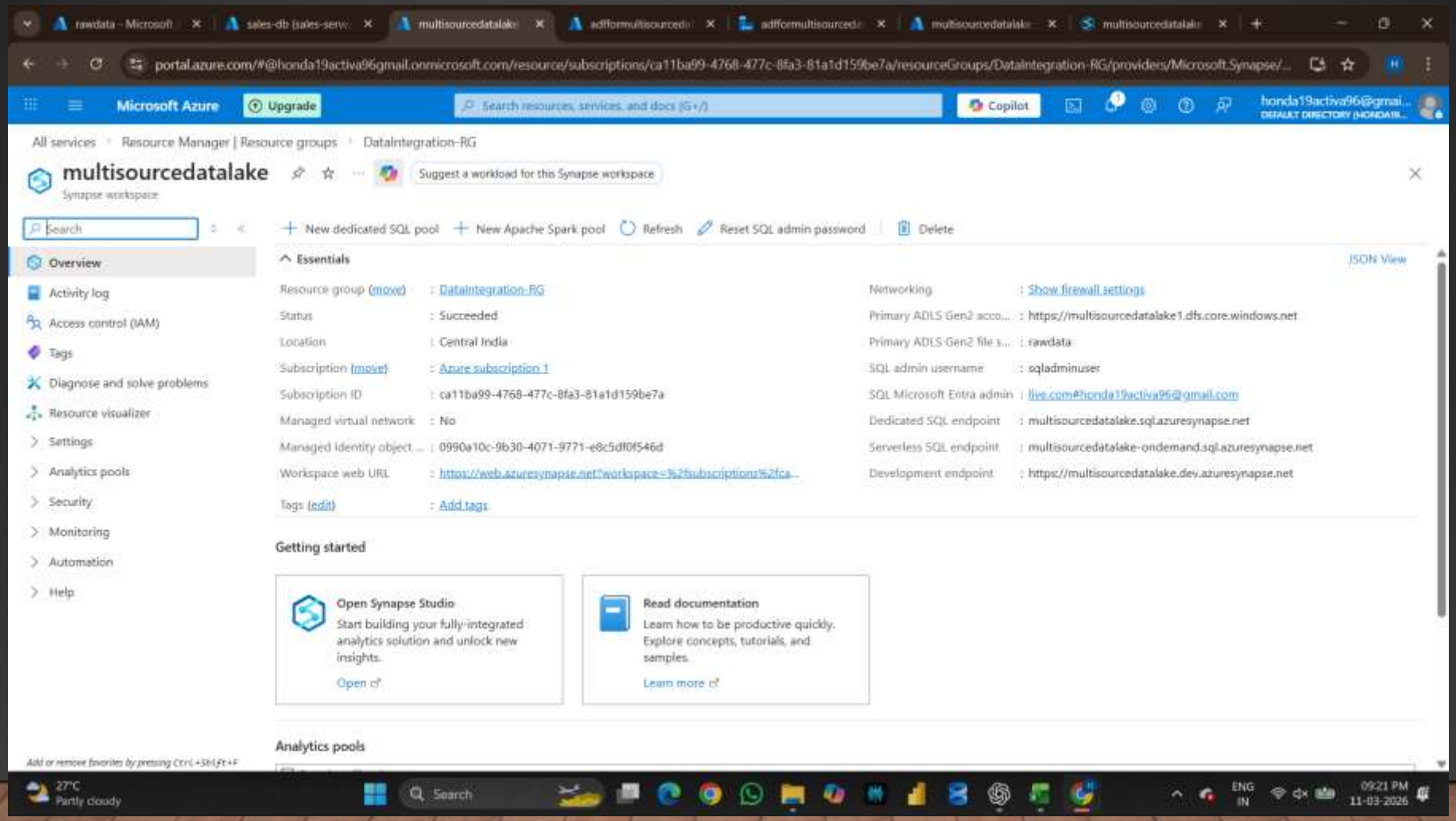


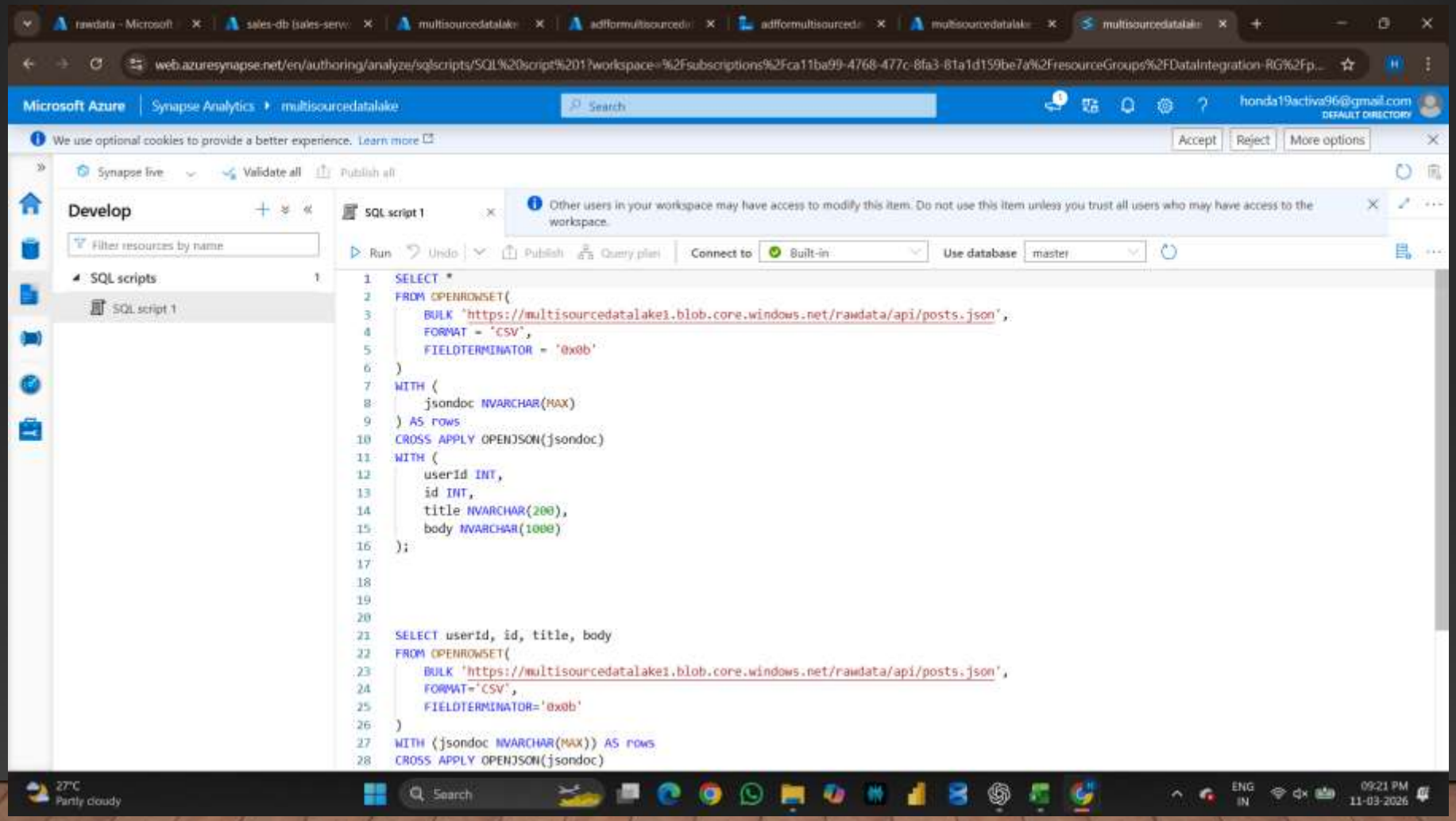
ADD PIPTLINE TRIGGER



AZURE SYNAPSE ANALYTICS



IN DEVELOP MODE WRITE A CODE



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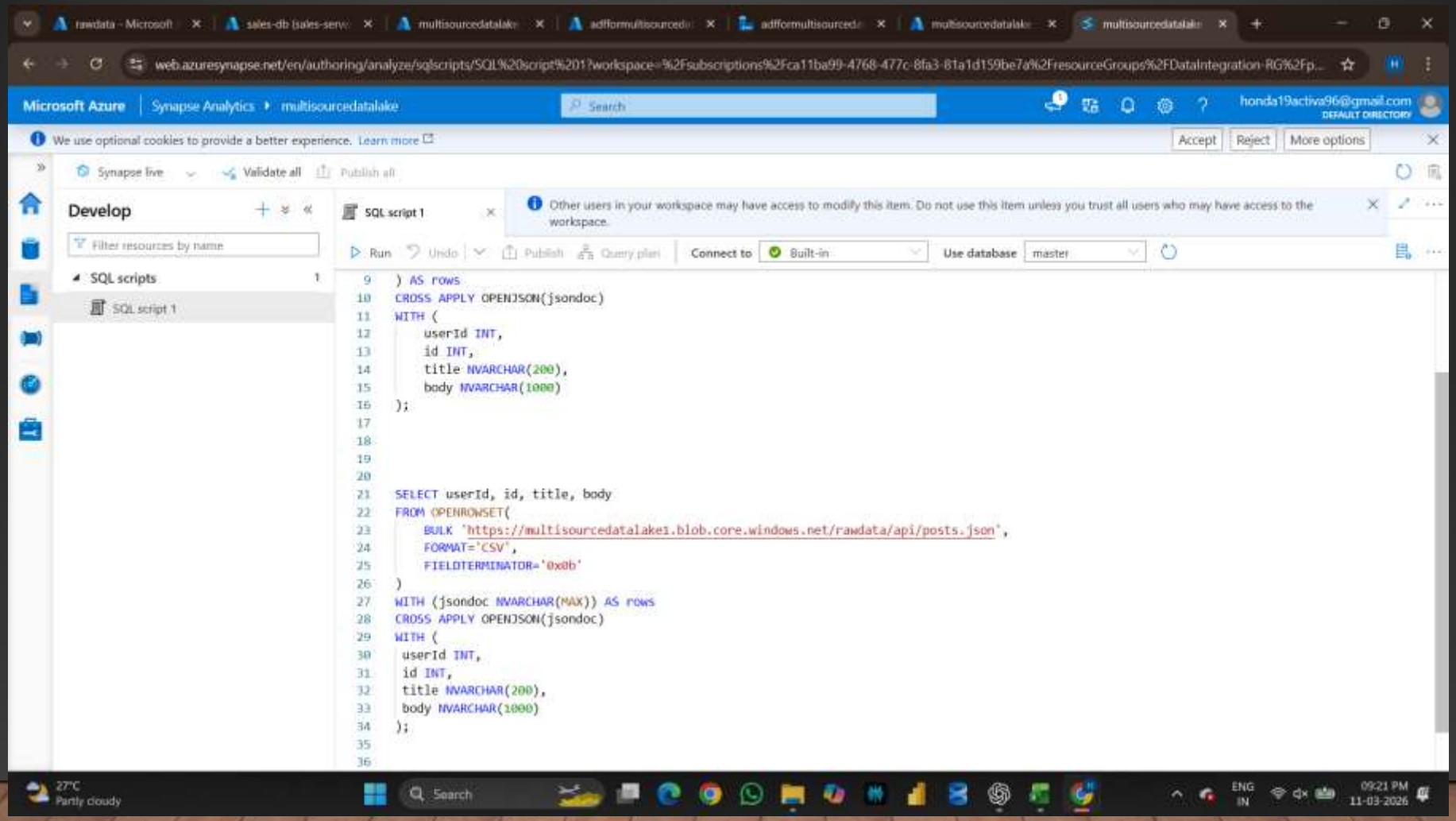
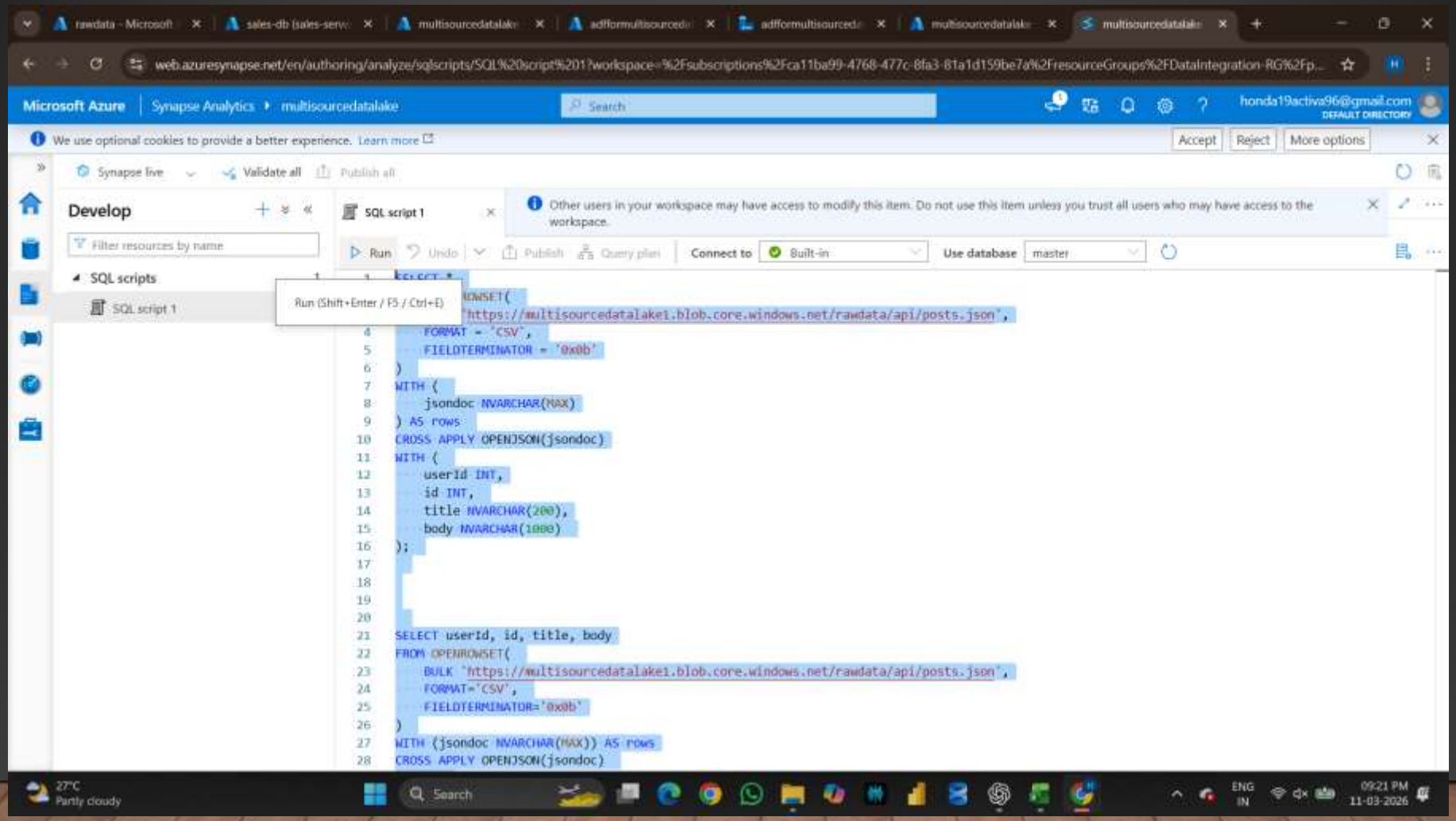
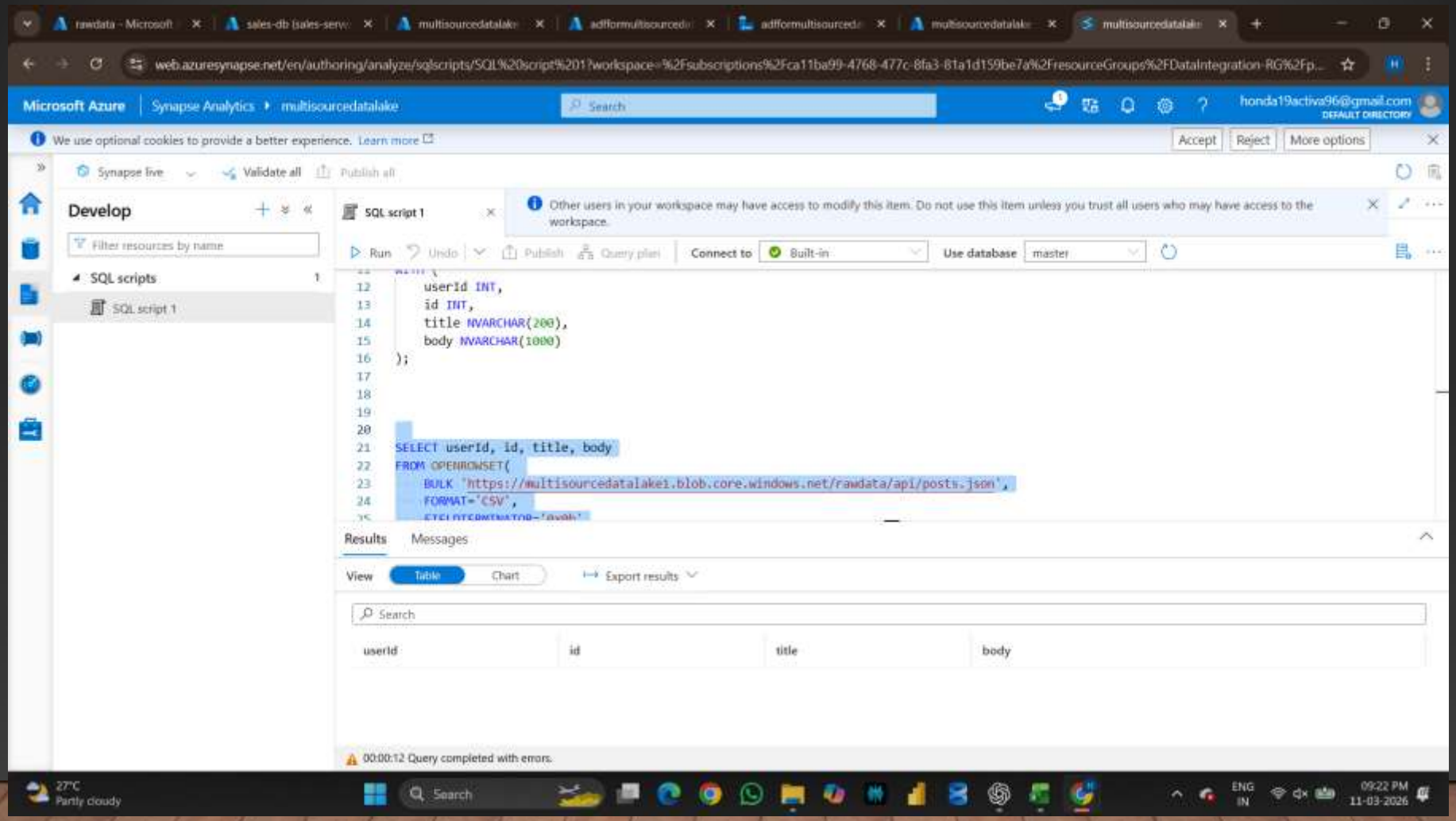


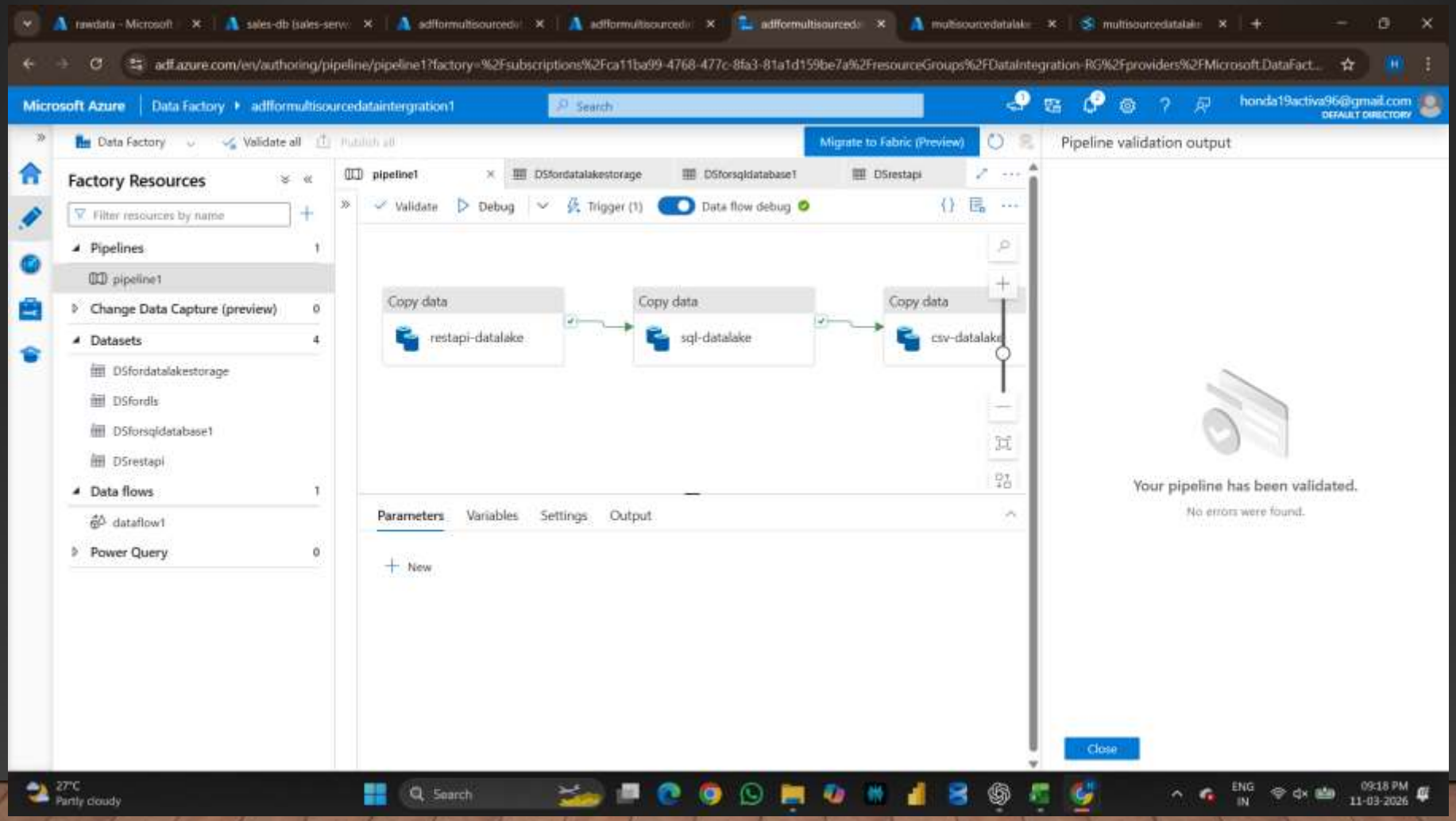
NOW RUN THE CODE



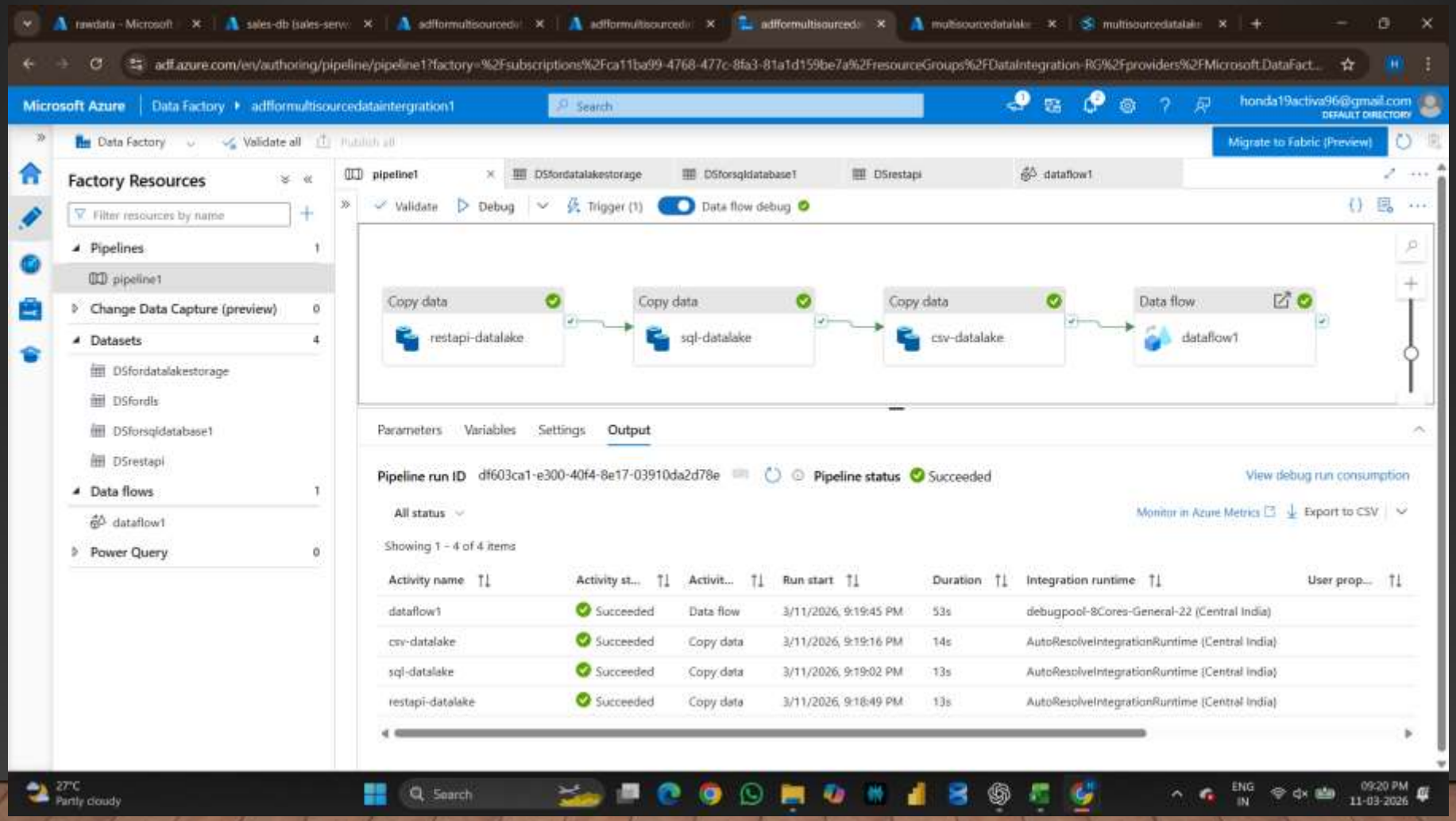
NOW RUN THE CODE



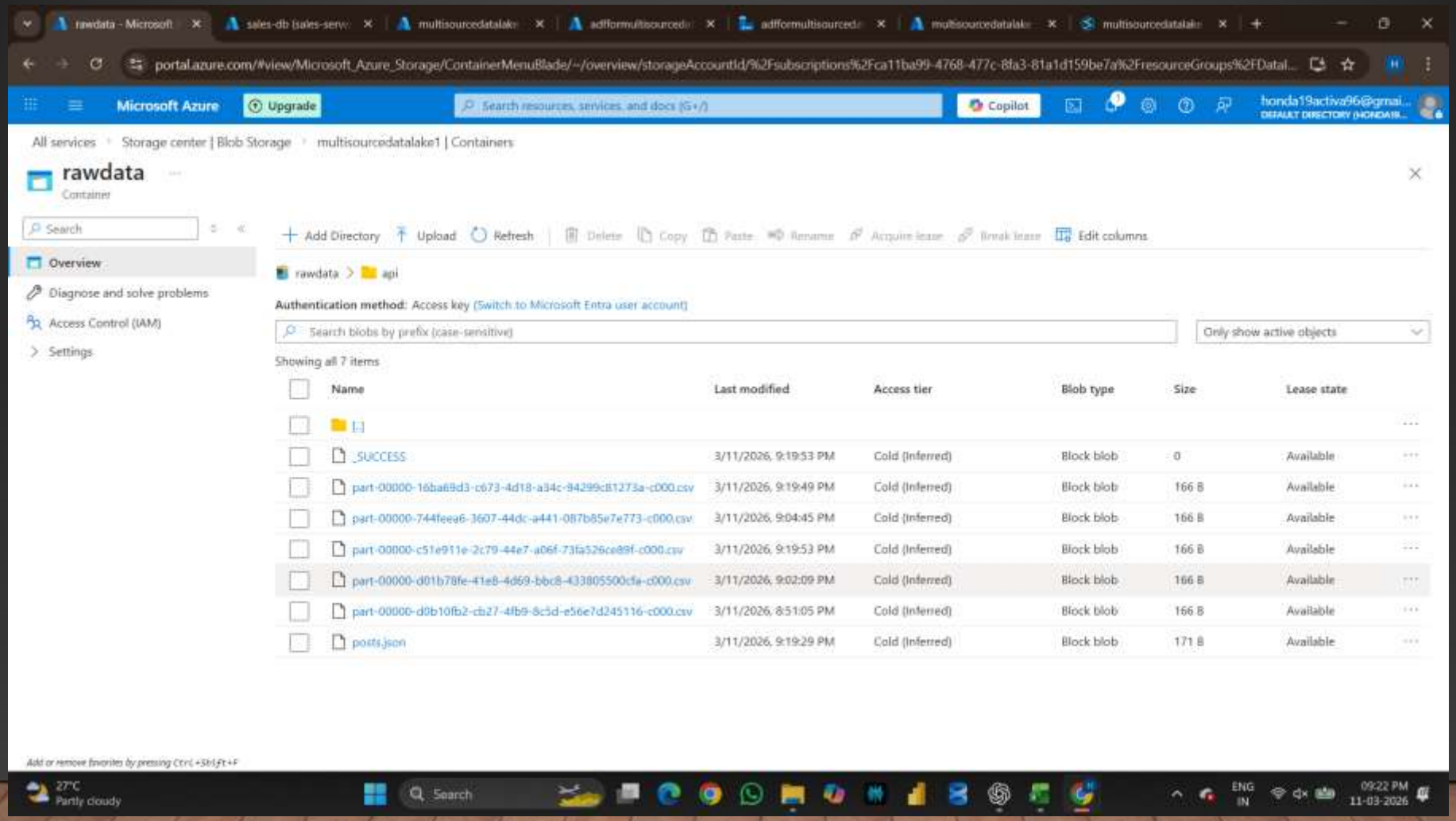
PUBLISH ALL & VALIDATE THE EVENTS



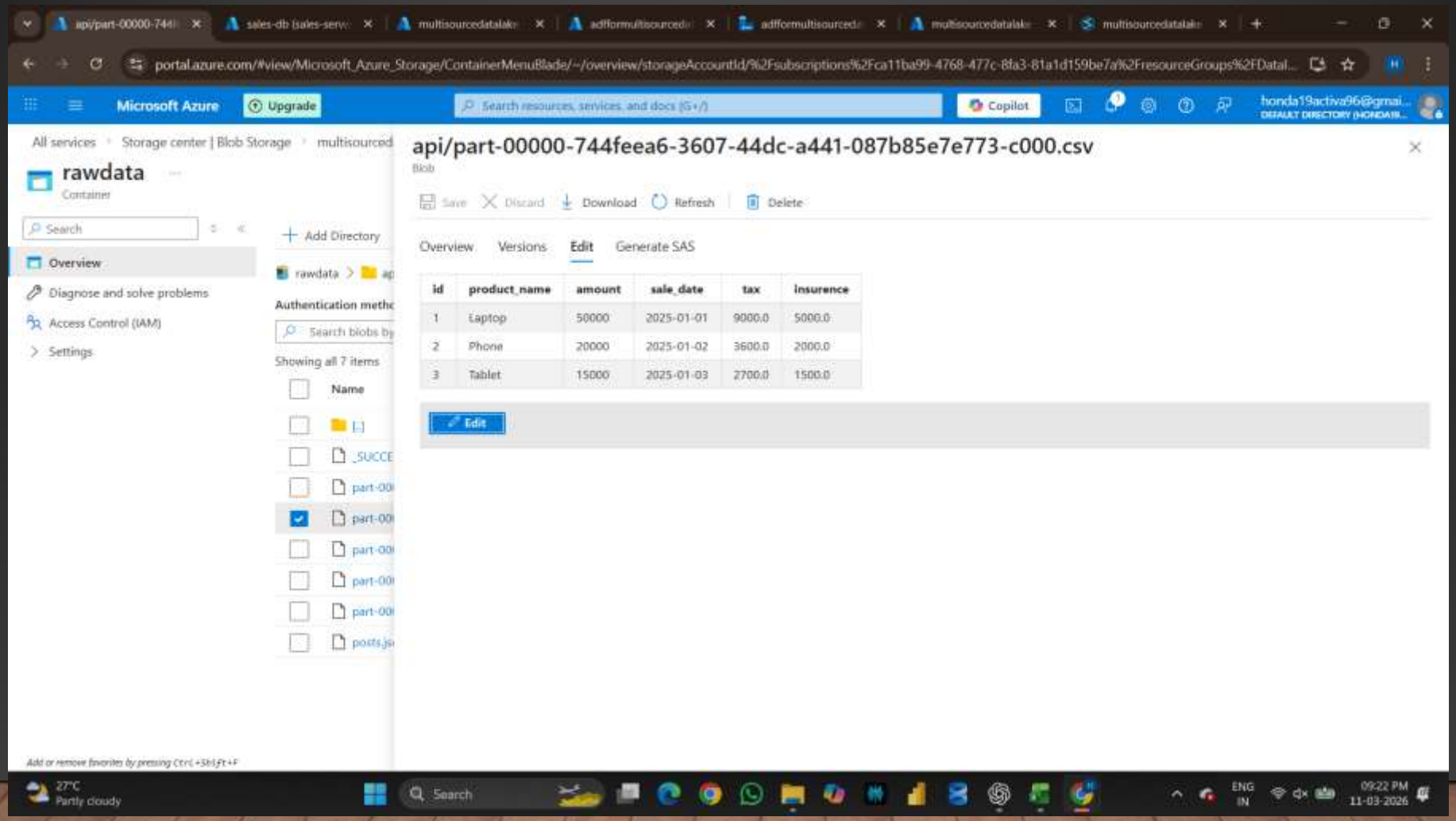
YOUR PIPELINE IS SUCCEEDED



NOW YOU CAN CHECK THE OUTPUT IN STORAGE CONTAINER



FINAL OUTPUT



THANKYOU

BY

S . HAREESWARAN